

Social Responsibility in Secondary Markets*

Marc Kaufmann
Central European University

Malte Kornemann
University of Bonn

Botond Kőszegi
University of Bonn

May 20, 2026

Abstract

We study how secondary markets for durable goods interact with consumers' social-responsibility motives to mitigate environmentally harmful new production. On the positive side, secondary markets may allow responsible consumers to acquire used goods that would otherwise be discarded, reducing premature waste. On the negative side, secondary markets introduce two major harmful forces. First, the possibility of buying used goods and thereby causing less harm can raise the demand of responsible consumers, often increasing the production necessary to serve the market. Second, said demand can increase the price of used goods, encouraging purchases of new goods. These forces imply that if used goods may have negative private consumption value, then secondary markets can enhance or erode the benefits of social responsibility. If, instead, used goods have positive private value, then secondary markets always (weakly) erode the benefits of social responsibility. Importantly, the logic underlying the harmful forces does not apply to new products sold after their production and associated externalities have been sunk. In the unique equilibrium, consumers treat such goods as having the same externality as production.

Keywords: socially responsible consumers, climate change, externalities, secondary markets, durable goods, used goods

JEL Codes: D01, D11, D50, D62, D64, D91

*We thank Peter Andre, Sarah Auster, Alexander Cappelen, Marcel Fafchamps, Mats Köster, Klaus Schmidt, Josh Schwartzstein, and conference and seminar audiences for comments. Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – EXC 2126/2 – 390838866. Supported by funds from the Austrian National Bank (Austrian Central Bank, Anniversary Fund, project number: 19068).

1 Introduction

Secondary markets for consumer durables are touted as an important tool for reducing environmentally harmful new production, making them a central component of emerging government regulations and initiatives. One potential mechanism for their environmental benefit is based on their well-known allocative role: transferring used items to higher-value consumers can raise the service flow from goods, reducing the demand for new production.

Intuition and discussions, however, suggest that secondary markets may reduce production through another channel as well: by encouraging socially responsible consumers — consumers who aim to reduce the externalities they cause — to forego otherwise appealing new products. Indeed, sustainability is one main motive that consumers cite for purchasing used products, and that second-hand retailers invoke in their marketing.¹ Furthermore, researchers and observers claim that outcomes improve as a result of substitution to second-hand goods (e.g., Sandin and Peters, 2018, Klooster et al., 2024). A tempting conclusion is that the prospect of socially responsible demand raises the value of secondary markets.

In this paper, we investigate the above intuitions regarding the positive interaction between secondary markets and social responsibility. This requires us to incorporate responsible consumers into a dynamic product market — a first in the literature — and to develop techniques for analyzing the equilibrium that obtains. Using our methodology, we identify settings in which the existing intuitions are correct, but also find an important class of situations in which they are backwards.

The effects hinge on three main forces. First, by providing a less harmful purchase option, used goods, secondary markets can raise the demand of responsible consumers, which in equilibrium is often met by increased production. Second, the preceding demand can raise the price of used goods, which increases resale values and thereby raises new purchases by less responsible consumers. Third, however, secondary markets may increase consumption of products that would otherwise be inefficiently discarded, reducing the need for new products.

Based on these forces, we provide a simple characterization: if used goods have positive private consumption utility, then secondary markets *always* (weakly) erode responsible behavior; and if

¹ Numerous papers investigate consumers’ reasons for buying second-hand products or participating in the sharing economy, with sustainability consistently being important (e.g., Guiot and Roux, 2010, Turunen and Leipämaa-Leskinen, 2015, Edbring et al., 2016, Hamari et al., 2016, Styvén and Mariani, 2020, Rodrigues et al., 2023). This is reflected, for instance, in the marketing of major second-hand clothing retailers Vinted, Poshmark, and ThredUp. Vinted emphasizes that second-hand clothing “is better than new for the climate” (<https://company.vinted.com/sustainability>). Poshmark wants to make “shopping and selling simple, social, and sustainable” (<https://poshmark.com>). And ThredUp writes that buying second-hand is about “standing for sustainability” (<https://www.thredup.com/about>). All accessed June 1, 2025.

used goods may have negative private consumption utility, then secondary markets can facilitate or erode responsible behavior. We also show that the beneficial case applies, after a reinterpretation, to a recycling system that is privately uneconomical but has sufficiently little material loss. Finally, we establish that the first harmful force does not operate for new goods whose production and associated externalities are sunk — in equilibrium, these goods are just as harmful as made-to-order goods.

We begin in Section 2 by introducing our model. In each period, consumers can buy new goods at a fixed price P , thereby causing a production externality, and trade used goods at market-determined prices, thereby not generating a direct externality. Between periods, all goods lose a portion f of their value through quality deterioration or breakage, and new goods become used, yielding an extra disutility of l per unit. Crucially, we assume (extending the static framework of Kaufmann et al., 2024) that beyond standard private consumption utility, a consumer has rational consequentialist social concerns. This means that (i) she derives disutility in proportion $k \geq 0$ to the rise in production she causes through her purchases; and (ii) her effect arises both directly from her purchase of new goods and indirectly from her (infinitesimal) effect on used prices, which can induce others to switch between new and used goods. Since the latter, indirect effect depends on the behavior of other responsible consumers, it is endogenously determined in equilibrium. Hence, we look for steady-state competitive equilibria in which consumers’ behavior as implied by (i) and their effects as derived from (ii) are mutually consistent. To isolate the interaction between secondary markets and social responsibility from standard allocative considerations, we assume throughout the main text that all consumers have the same private consumption utility, but differ in their “social coefficients” k .

Because the general framework is challenging to analyze, we start by developing our main insights in a special case with two restrictions. First, we impose that the distribution of social coefficients k is binary, with some consumers being selfish ($k = 0$) and some being responsible ($k = \kappa > 0$). Second, we focus on “strict” equilibria, in which both consumer types have a strict preference between new and used products. Then, an infinitesimal price change does not induce anyone to switch between new and used products, so purchases have no indirect effects on production. Further, consistency requires that with only new purchases having an effect on production, both types of consumers indeed have strict preferences between the products.

We analyze our basic model in Section 3. We first consider situations in which the distaste l for used goods is modest ($l < (1 - f)P$), so that even without a secondary market, discarding a used

item and buying new instead is privately suboptimal. This positive-value condition for used goods might, for instance, apply to higher-quality apparel, whose resale markets have grown tremendously in the last few years. We show that in any strict equilibrium, used goods are relatively expensive (albeit cheaper than new goods), and in each period selfish types buy new goods and sell their used goods to responsible types. Because used goods do not generate an externality, responsible types purchase more lavishly than without a secondary market, and the high resale price encourages selfish types to do the same, yielding higher production and lower welfare. From an economics perspective, this provides a new explanation for why sellers may benefit from secondary markets. From a social perspective, it suggests that resale programs can even serve as a form of greenwashing — a strategy that appears environmentally responsible but in fact increases production and externalities.

Now suppose that the distaste for used goods is more substantial $((1-f)P < l < (1-f)(P+\kappa))$, so that a used item has negative private value. This condition might, for instance, apply to low-quality fast fashion and similar merchandise commonly found at thrift stores. If the share of responsible consumers is sufficiently low, then the secondary market allows selfish types to “sell” some of their (otherwise discarded) used goods at a price of zero to responsible types, who forego new goods as a result. Hence, the secondary market reduces production and waste, increasing welfare. For a higher share of responsible consumers, however, used goods sell at a positive price, encouraging selfish consumers to buy more new goods. Consequently, the welfare effect of the secondary market is now ambiguous.

Through a reinterpretation, this version of our model applies to a recycling system. Suppose that l is not a preference parameter, but the cost of converting a used product into $1-f$ units of a privately equivalent recycled product. Then, $l > (1-f)P$ means that recycling is privately not economical. Still, responsible consumers are willing to buy recycled goods, and if the loss from recycling is sufficiently small (f is low), then the system lowers production and raises welfare.

Finally, if the private distaste for used goods is very large ($l > P + \kappa$), then all consumers discard used products in every period. Hence, the secondary market must remain dormant and has no effect on production.

In Section 4, we demonstrate the robustness of our main insights by allowing for a general distribution of social coefficients k and investigating not just strict equilibria. The technically difficult case is when the distaste l for used goods is low, which for notational simplicity we analyze by setting $l = 0$. Unlike in strict equilibria, a consumer may now impact production today and in all future periods, so her total impact is no longer immediate. In particular, there is a contemporaneous

cross-market effect: used-good consumption can indirectly increase production today by raising the used price and thereby inducing others to buy new. In addition, there is an intertemporal effect: directly or indirectly increasing production today raises used supply in the future, which can indirectly affect production through the used price in future periods.

To obtain traction, we reformulate our equilibrium conditions in terms of the above contemporaneous cross-market effect. Using this reformulation, we establish that the secondary market always weakly erodes responsible behavior. Specifically, (i) there is always an equilibrium in which production and welfare are as without the secondary market; and (ii) there is often also an equilibrium in which production is strictly higher and welfare is strictly lower than without the secondary market. The logic of a type-(ii) equilibrium is similar to that of a strict equilibrium above: buying used products is less harmful than buying new ones, so ultimately responsible as well as less responsible consumers purchase more lavishly than without the secondary market. In the type-(i) equilibrium, however, everyone is indifferent between new and used goods, and buying either has the same effect on production, so the secondary market does not spur consumption. We show that (fixing other primitives) the set of distributions of social coefficients k for which the harmful type-(ii) equilibrium exists is in a sense dense.

Beyond strengthening our conclusions, the general model yields further economically relevant points. As a theoretically interesting possibility, we note that an increase in the distribution of consumers' social coefficients k (in first-order stochastic dominance) can greatly lower welfare by introducing an equilibrium of type (ii).

In addition, we evaluate the common argument that purchasing new retail goods whose production has already occurred — and whose externalities have therefore already been generated — is harmless. The reason behind the potential harmfulness of secondary markets, that used products are less harmful than new ones, may seem to support this reasoning. In contrast, we show that under weak conditions, buying a previously produced item is equivalent to buying from the manufacturer. This is because the unique equilibrium with recently-produced new goods being sold is of type (i). Hence, among rational consumers at least, the common argument is flawed.

We conclude in Section 5 by highlighting areas for future research, including policy interventions, endogenous product durability, and responsible consumers' efforts to alter the beliefs of firms or policymakers. The Appendix contains proofs and the analysis of some additional cases. We show that many implications are similar if responsible consumers are naive in that they do not understand their indirect effects, or deontological in that they care only about their own action or direct

effect. In a strict equilibrium, in particular, such consumers behave the same way as rational consequentialist consumers. And we consider one setting in which a classical allocative role for the secondary market is also present, for instance because high-income fashion seekers who dislike used goods interact with low-income consumers who care primarily about the price.

Policy relevance and related literature Our analysis informs recent policy proposals and countless discussions in the business and popular press on secondary markets. For example, the European Union’s (EU’s) Circular Economy Action Plan of 2020, a comprehensive agenda to help achieve carbon neutrality, aims to engender widely available second-hand markets. One specific method is the creation “Digital Product Passports” containing information on the product’s history, which may help engender trust in buyers. Similarly, California’s Responsible Textile Recovery Act of 2024 mandates sellers to collect unwanted textiles and encourage second-hand and thrift stores.² Consistent with such steps, a McKinsey report by Gatzer et al. (2022) predicts that the EU market for used and recycled consumer goods will rise to €400-650 billion (22-38% of the total consumer-goods market) by 2030. Our findings suggest a potential for unintended consequences due to an interaction with consumers’ social-responsibility motives.

The results we find may also be helpful for individual consumers thinking about their impacts from new and used purchases when a secondary market is already in place. On the one hand, many consumers and economists have made the above argument that purchasing new products from retailers is harmless. Our framework helps clear up this confusion. On the other hand, some audience members and friends have reacted to our results by expressing guilt about used purchases. Such reactions are unjustified: in all equilibria, buying used has a weakly lower externality impact than buying new, so it is the more responsible thing to do. This is the case even if establishing the secondary market is socially harmful, so that advocating for it is not a good idea.

Within economics, our paper contributes primarily to the theoretical analysis of social responsibility among consumers and to the understanding of durable goods and secondary markets. To the best of our knowledge, we are the first to combine these topics, as well as the first to analyze the effects of responsible consumers in a dynamic product market.

For modeling markets with socially responsible consumers, we build on the static framework

² See, e.g., <https://www.eumonitor.eu/9353000/1/j9vvik7m1c3gyxp/vkww8lwipmzm> and https://environment.ec.europa.eu/strategy/textiles-strategy_en on the Circular Economy Action Plan; https://en.wikipedia.org/wiki/EU_Digital_Product_Passport and [https://www.europarl.europa.eu/RegData/etudes/STUD/2024/757808/EPRS_STU\(2024\)757808_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2024/757808/EPRS_STU(2024)757808_EN.pdf) on Digital Product Passports; and https://leginfo.ca.gov/faces/billTextClient.xhtml?bill_id=202320240SB707 on the Responsible Textile Recovery Act. All accessed August 14, 2025.

of Kaufmann et al. (2024), but our dynamic setting leads to different theoretical challenges and economic mechanisms.³ In much of the other research on the market effects of responsible consumers and investors (Sobel, 2007, Dufwenberg et al., 2011, Pástor et al., 2021, Piccolo et al., 2022, Aghion et al., 2023, Arnold, 2023, Dewatripont and Tirole, 2024), a person’s social concern depends exogenously on actions or outcomes, whereas in our setting it depends on the consumer’s endogenous equilibrium impact. Furthermore, papers that do consider impact-based social preferences (e.g., Norwood and Lusk, 2011, Moisson, 2020, Green and Roth, 2021, Hakenes and Schliephake, forthcoming, Broccardo et al., 2022, Herweg and Schmidt, 2022, Krahnen et al., 2023, Trammell, 2023, Oehmke and Opp, 2025) study questions and use methods that are different from ours.

There is also an extensive body of classical theory on durable goods and secondary markets. This literature investigates questions such as the choice of durability by firms (e.g., Swan, 1970), time inconsistency (e.g., Coase, 1972), planned obsolescence (e.g., Bulow, 1982), and monopolists’ incentives to interfere with secondary markets (e.g., Hendel and Lizzeri, 1999, 2002), but does not consider markets with socially responsible consumers.

On the empirical side, there is substantial evidence for the type of consumer we analyze. Many studies, including incentivized experiments by Meier et al. (2023), Rodemeier (2023), Schulze Tilling (2024), and Andre et al. (forthcoming), show that consumers care about their externality effects. The importance of social concerns in the purchase of second-hand goods is also well-documented in the literatures on sustainability (e.g., Borusiak et al., 2020, Varah et al., 2021, Rodrigues et al., 2023) and the sharing economy (e.g., Hamari et al., 2016).

2 Basic Framework

Environment We consider an infinite-period economy with markets for new and used goods each period. New goods are available in perfectly elastic supply at price P , and their sale raises externality-generating production one-to-one. As a new good ages one period, it becomes used. As a used good ages one period, it loses a portion f of its value, so that one unit of a one-period-older good is equivalent to $1 - f$ units of a one-period-younger good. We measure the quantities of all used vintages in units of one-period-old products, allowing us to collapse used consumption into a

³ In fact, by assuming that new products are available in fully elastic supply, we abstract from the main effect on which Kaufmann et al.’s predictions rely, “dampening”. Dampening means that when a consumer raises her consumption, she raises the market price and thus induces others to consume less, lowering the externality she brings about. This static effect does not appear to interact with the dynamic issues due to durability and secondary markets that we investigate in this paper.

single number. Accordingly, if in one period a person consumes amounts $c_n \geq 0$ and $c_u \geq 0$ of the new and used goods, respectively, then she starts the next period with an amount $c_n + (1 - f)c_u$ of used goods. Used goods can be bought and sold at a market-determined price p_u , and such trade does not generate direct externalities. There is free disposal, so that the used price p_u must be non-negative.

Preferences and equilibrium We define a *steady-state competitive equilibrium* as a situation satisfying five conditions. The first three pertain to outcomes and expectations. First, there is a constant used price $p_u^* \geq 0$ as well as constant consumption. Second, p_u^* balances the secondary market, i.e., the market features either $p_u^* = 0$ and excess supply or market clearing in each period. Third, a consumer treats a surprise deviation in the used price from p_u^* as idiosyncratic, expecting a return to p_u^* in the future.

The fourth condition defines individual behavior. In each period, a consumer observes the current used price $p_u \geq 0$, and chooses consumption amounts $c_n \geq 0$ and $c_u \geq 0$ to maximize

$$U_k(c_n, c_u) = \underbrace{u(c_n + (1 - f)c_u) - lc_u}_{\text{gross consumption utility}} - \underbrace{Pc_n - p_u c_u}_{\text{current payments}} + \underbrace{p_u^*(c_n + (1 - f)c_u)}_{\text{expected resale value}} - \underbrace{k \cdot (c_n r_n^* + c_u r_u^*)}_{\text{social concern}}. \quad (1)$$

Gross utility depends on “total consumption” $c_n + (1 - f)c_u$, so that — similarly to the situation with older and younger used goods — one unit of a used good is substitutable with $1 - f$ units of a new good. In addition, the consumer may have a further distaste $l \geq 0$ for used goods. The function $u(\cdot)$ is twice continuously differentiable and strictly concave with $\lim_{c \rightarrow 0} u'(c) = \infty$ and $\lim_{c \rightarrow \infty} u'(c) = 0$. The next terms are the payments for current consumption and the expected secondary-market value of used goods next period. To these elements of standard private utility, the last term adds social concerns. A consumer’s concern is proportional to her social coefficient k , which in our basic model is binary: a share g of consumers is selfish ($k = 0$), and a share $1 - g$ is socially responsible with $k = \kappa > 0$. Furthermore, the consumer’s concern derives from her equilibrium impact $c_n r_n^* + c_u r_u^*$ on total production over time, where r_n^* and r_u^* denote the impacts of new and used consumption, respectively.

As the fifth condition for equilibrium, we impose a consistency requirement on r_n^* and r_u^* by adapting Kaufmann et al.’s (2024) definition of competitive equilibrium with small rational consequentialist responsible consumers. An individual can affect production directly by consuming new, and indirectly by changing used prices and thereby inducing others to switch between new and

used consumption. Kaufmann et al. show that such indirect effects on quantities are in general not negligible, even though a small consumer’s effect on prices is. Consistency means that if consumers expect to have impacts r_n^* and r_u^* , then the effects of new and used consumption — which, due to the indirect effects, depend on demand and hence on r_n^* and r_u^* — indeed aggregate to r_n^* and r_u^* .

Because the general definition and analysis of consistency are cumbersome, we defer it to Section 4. For our basic model, we focus on “strict” equilibria, in which both selfish and responsible consumers have a strict preference between new and used products. Then, there are no indirect effects of consumption — a marginal change in used prices does not induce anyone to switch between the two products — so $r_n^* = 1$ and $r_u^* = 0$. In addition, consistency requires that consumers maximizing (1) with $r_n^* = 1$ and $r_u^* = 0$ have strict preferences for both $k = 0$ and $k = \kappa$.

Comments (1) Although we have assumed that used goods lose a share f of their value between periods, an equivalent model arises if a share f breaks, and the rest retain full value; or a total depreciation of f results from a combination of value loss and breakage. The assumption that depreciation is geometric is necessary for tractability. (2) We use the anchored-belief assumption — that a consumer takes deviations in the used price p_u from the steady state as one-time — only for vanishingly small price deviations that a single consumer can cause. This is justified because noise in price determination (which we do not model but is present in reality) makes small deviations from equilibrium undetectable.⁴ (3) While U_k is defined over current consumption only, maximizing intertemporal utility with discount factor δ approaching 1 is in the limit equivalent to maximizing U_k . This is because available consumption choices, their ranking, and (hence) optimal behavior in a period are independent of past or future consumption. (4) We assume exogenously that buying a new good increases production by one unit. Such an assumption corresponds most closely to a system of made-to-order manufacturing. We consider the more realistic situation where goods are sold after production in Section 4.3. (5) Consumers in our theory are rational and consequentialist. Appendix A shows that our main insights, including all insights from the basic model, extend to other definitions of social responsibility.

⁴ Our specification implicitly also imposes that upon deviation from equilibrium, a consumer still expects to have the equilibrium impacts r_n^* and r_u^* . Such an assumption is natural for consumers with anchored beliefs and a negligible price impact.

Benchmark We will compare outcomes with a secondary market to those without one. In the latter case, a consumer’s steady-state consumption c_n^*, c_u^* solves

$$(c_n^*, c_u^*) = \arg \max_{\substack{c_n, c_u: \\ c_u \leq (c_n^* + (1-f)c_u^*)}} u(c_n + (1-f)c_u) - lc_u - Pc_n + p_u^*(k)(c_n + (1-f)c_u) - k \cdot c_n. \quad (2)$$

The absence of trade in used goods introduces three differences relative to (1): the term $-p_u c_u$ is absent, there are no indirect effects of consumption ($r_n^* = 1, r_u^* = 0$), and used consumption c_u is bounded by the amount of used goods available from before. Further, we impose that used goods have a shadow value of $p_u^*(k) = \max\{(1-f)(P+k) - l, 0\}$ in the next period. The first argument reflects that having one unit of the used good, rather than buying $1-f$ units of the new one, saves $(1-f)(P+k)$ in financial and social-concern costs next period, but it also lowers utility by l . The second argument reflects free disposal. Introducing a shadow value to account for these considerations is necessary because (2) is again defined over current consumption only. The same shadow value would emerge in the alternative specification with discounting mentioned earlier.

Welfare and waste We assume that production has an exogenously given externality cost of $K > 0$ per unit. Hence, we define steady-state social welfare as per-period total gross consumption utility minus $(P+K)$ times per-period production. Moreover, we posit that $k < K$ and $l < (1-f)(P+K)$. The first inequality means that no consumer fully internalizes the social cost of the externality she causes. The second inequality implies that disposing of used goods and buying new goods instead is socially inefficient, i.e., it creates “premature waste.” To see this, notice that replacing a unit of the used good with $1-f$ units of the new good eliminates a consumption disutility of l , but also has a social cost of $(1-f)(P+K)$.

For interpreting results, it is useful to distinguish premature waste from “unavoidable waste,” which is socially inefficient to consume ($l > (1-f)(P+K)$). As we have mentioned, our framework accommodates the possibility that some used goods break each period. We can naturally interpret such broken items — which must be discarded — as unavoidable waste. The extent to which real-life waste is premature or unavoidable is unclear, so the existence of waste is consistent with all versions of our model below.⁵

⁵ Relatedly, while we focus on production externalities, any externalities from waste (e.g., environmental degradation from landfilling unwanted textiles) can be included in K . Since all new production eventually becomes waste when the products break, in steady state the two must equal in quantity.

Homogeneity in consumption utility In our model, all consumers have the same consumption-utility function. If consumers had different inherent tastes, there would potentially be trade on the secondary market for purely private reasons. We abstract from this classical force to isolate effects due solely to social responsibility. Since with only selfish consumers ($g = 1$) the secondary market has no impact on outcomes, its welfare impact stems from how it influences responsible consumers; and conversely, the same impact reflects how responsible consumers influence the value of the secondary market. These interactions are impossible to cleanly isolate when there is also trade for purely private reasons, as social responsibility necessarily affects these trades too. Nevertheless, in Appendix B we briefly discuss a model with heterogeneity in consumption utility.

3 Secondary Markets and Responsible Behavior

In this section, we characterize all strict equilibria, thereby allowing us to identify when secondary markets facilitate versus erode socially responsible behavior. As one practical illustration, we use apparel markets; other potential applications include electronics, electric vehicles, furniture, books, and sports or outdoor equipment. We consider three cases.

3.1 Trading Privately Valuable Goods

We first suppose that $l < (1 - f)P$. The condition means that in the absence of a secondary market, discarding used goods to buy new goods is privately suboptimal. Indeed, replacing a unit of the former with $1 - f$ units of the latter eliminates a consumption disutility of l , but also costs $(1 - f)P$. Such privately valuable used goods appear to dominate many resale markets, for example in used apparel and mobile phones — sectors growing much faster than traditional donation and thrift stores.⁶ Without the opportunity to resell, brand-name clothing may be worth keeping even if rarely worn, and an old mobile phone may be worth having as a backup.

Proposition 1 (Erosion of Responsible Behavior). *There are $\underline{g}$ and $\bar{g}$ satisfying $0 \leq \underline{g} < \bar{g} < 1$ such that a strict equilibrium exists if and only if $\underline{g} < g < \bar{g}$. Any strict equilibrium features strictly higher total consumption by both consumer types, strictly higher production, and strictly lower social welfare, than without the secondary market. The used price satisfies $(1 - f)P - l < p_u^* < \min \{P, (1 - f)(P + \kappa) - l\}$.*

⁶ See, for instance, ThreadUp (2024). The US apparel resale sector grew from \$3 billion in 2017 to \$22 billion in 2022, and is expected to reach \$42 billion by 2027. The same numbers for traditional donation and thrift stores are \$17 billion, \$22 billion, and \$28 billion, respectively.

Proposition 1 says that in the current case, a secondary market can only erode responsible behavior. We sketch the argument in the notationally simple case of $l = 0$.

Idea of proof ($l = 0$) When $l = 0$, consumption utility depends only on total consumption $c = c_n + (1 - f)c_u$. Equation (2) implies that without a secondary market, type $k \in \{0, \kappa\}$ chooses total consumption $c_k^{\neg sm}$ to solve $u'(c_k^{\neg sm}) - (P + k) + (1 - f)(P + k) = 0$, or $u'(c_k^{\neg sm}) = f(P + k)$. Further, a consumer's steady-state consumption amounts must satisfy $c_u = c_n + (1 - f)c_u = c$, so $c_n = fc_u = fc$; intuitively, new purchases compensate for the depreciation f of products. Hence, total new consumption, which equals new production, is $f(gc_0^{\neg sm} + (1 - g)c_\kappa^{\neg sm})$ in each period.

Equation (1) implies that in the presence of a secondary market, a necessary and sufficient condition for the consistency of $r_n^* = 1$ and $r_u^* = 0$ — that selfish types strictly prefer new goods and responsible types strictly prefer used goods — is $(1 - f)P < p_u^* < (1 - f)(P + \kappa)$. Further, types $k = 0$ and $k = \kappa$ choose total consumption amounts c_0^* and c_κ^* to solve

$$u'(c_0^*) = P - p_u^* \quad \text{and} \quad u'(c_\kappa^*) = \frac{f}{1 - f} \cdot p_u^*, \quad (3)$$

respectively. The equilibrium exists if g is such that the market-clearing condition $gc_0^* = (1 - g)fc_\kappa^*/(1 - f)$ holds, i.e., the amount selfish consumers sell each period — everything they bought new last period — equals the amount socially responsible consumers need to compensate for depreciation (since c_κ^* is responsible types' total consumption, their used consumption is $c_\kappa^*/(1 - f)$).

In that case, production per period is $gc_0^* = fgc_0^* + (1 - f)gc_0^* = fgc_0^* + f(1 - g)c_\kappa^* = f(gc_0^* + (1 - g)c_\kappa^*)$. Now $(1 - f)P < p_u^* < (1 - f)(P + \kappa)$ implies that $P - p_u^* < fP$ and $\frac{f}{1 - f} \cdot p_u^* < f(P + \kappa)$, so $c_0^* > c_0^{\neg sm}$ and $c_\kappa^* > c_\kappa^{\neg sm}$. As a result, both types' total consumption and production are strictly higher than without the secondary market. $\square$

Intuitively, responsible consumers understand that by buying used goods, they are not generating an externality — they merely crowd out used consumption by others. This liberates them to consume a lot, with their increased demand also driving up the used price p_u^* . Anticipating a high resale price in turn induces selfish consumers to consume a lot as well. Importantly, the latter is an equilibrium feedback effect that responsible consumers — unable to influence expectations regarding *future* prices — cannot mitigate by changing their current behavior. Since all purchases ultimately come from new production, the lavishness of everyone's consumption is harmful for the level of the externality and social welfare.

In some situations, the secondary market not only raises the harm that stems from the externality, but also lowers the population's average *private* consumption utility net of prices paid. As a notable extreme example, suppose that $l = 0$ and $p_u^* \approx (1 - f)(P + \kappa)$. Then, responsible types' consumption utility is approximately the same as without the secondary market, while selfish types' consumption utility is strictly higher. But since $u'(c_0^{-sm}) = fP$, the latter increase is in the range where marginal utility is below fP , the per-period price of maintaining a unit of steady-state total consumption. Intuitively, the secondary market does nothing but act as a subsidy to selfish types that is inefficient even ignoring externalities.

While we have assumed perfectly elastic supply, a further point emerges with an increasing supply curve. Then, the secondary market may raise the new price, benefiting sellers.⁷ This possibility provides a novel argument for why sellers may like secondary markets, and is consistent with the fast-growing tendency of brand-name apparel makers to embrace resale.⁸ It also suggests that the promotion of resale may be a greenwashing strategy. Since in equilibrium used goods generate less of an externality than new goods, a firm with a resale program appears to be environmentally friendly. Yet having a secondary market raises new sales and is environmentally unfriendly.

3.2 Reducing Premature Waste

Next, we assume that $(1 - f)P < l < (1 - f)(P + \kappa)$, so that it is privately optimal to dispose of used goods and replace them with new goods in proportion $(1 - f)$. This assumption may describe fast fashion and other types of low-quality merchandise that become an unexciting nuisance after a short period of use.

Proposition 2 (Fragile Facilitation of Responsible Behavior).

- A. [Equilibrium Characterization.] *There are $\underline{g}, \bar{g}$ satisfying $0 \leq \underline{g} < \bar{g} < 1$ such that:*
- I. *If $g > \bar{g}$, then there is a unique strict equilibrium. In this equilibrium, $p_u^* = 0$, selfish types' total consumption is the same as without the secondary market and responsible types' total consumption is strictly higher, but the latter do not buy new products. Production is strictly*

⁷ E.g., suppose that the supply curve is a step function: it equals $P' < P$ up to quantity $q' < gc_0^*$, and P for higher quantities. This leaves the preceding equilibrium, including the new-good price P , unchanged. But if q' and P' are sufficiently close to gc_0^* and P , respectively, then the new-good price without a secondary market is P' .

⁸ E.g., ThreadUp (2024), or Bhattarai ("Old clothes, new customers," Washington Post, January 31, 2020). The number of brands with resale programs increased from 4 in 2018 to 163 in 2023. Common intuition and some research (e.g., Rust, 1986, Waldman, 1996) suggests that a secondary market harms sellers by lowering demand for new products. Swan's independence result (Swan, 1970, Sieper and Swan, 1973) implies that a secondary market has no effect on sellers. Other research shows that a secondary market can benefit sellers by allowing them to expand the customer base or price discriminate (e.g., Anderson and Ginsburgh, 1994, Hendel and Lizzeri, 1999).

lower and social welfare is strictly higher than without the secondary market.

II. If $\underline{g} < g < \bar{g}$, then there is a unique strict equilibrium. In this equilibrium, $p_u^* \in (0, (1 - f)(P + \kappa) - l)$ and both types' total consumption is strictly higher than without the secondary market, but responsible types do not buy new products.

III. If $g < \underline{g}$, then a strict equilibrium does not exist.

B. [Fragility of Welfare Gains.] For any P, κ, l, K, g , and f with $f^2(1 - g) > g(1 - f)$, there is a utility function $u(\cdot)$ such that in the unique strict equilibrium, total private consumption utility is strictly lower and production is strictly higher than without the secondary market.

The proposition says that for privately less valuable products, secondary markets can facilitate socially responsible behavior, albeit this beneficial effect is fragile. To begin, Part A characterizes strict-equilibrium outcomes. We sketch the argument for the most novel case, A.I., when the secondary market improves outcomes.

Idea of proof, Part A.I. Suppose first that there is no secondary market. Since $l > (1 - f)P$, selfish consumers strictly prefer new goods for all of their consumption each period, and face a shadow price of zero. This implies that they always toss their used goods from before. Since $l < (1 - f)(P + \kappa)$, however, responsible consumers strictly prefer used goods they already have to new goods. Nevertheless, they do buy new goods each period to compensate for depreciation.

In the presence of a secondary market, consider a candidate equilibrium with $p_u^* = 0$. Setting $r_n^* = 1$ and $r_u^* = 0$, Equation (1) implies that, consistent with a strict equilibrium, selfish consumers strictly prefer new goods, and responsible consumers strictly prefer used goods. Now since selfish types face the same p_u^* as without the secondary market, they consume the same amount. With responsible types shunning new production, therefore, the secondary market lowers total production. Such an equilibrium exists if the share of selfish consumers, g , is sufficiently high for their supply of used goods to exceed the amount responsible consumers need to compensate for depreciation. $\square$

Intuitively, a responsible consumer knows that if she buys a used product, she just reduces what selfish consumers discard, and does not affect current or future production. Hence, she makes a private sacrifice to decrease premature waste and thereby lower the need for production. By facilitating this sacrifice, the secondary market raises welfare. But because $p_u^* = 0$ and market clearing fails, the exchange of used goods is more akin to donation than to market trade.

The above equilibrium is potentially consistent with the reality of apparel donations to thrift stores and other organizations, which are typically of the low-quality type. Anecdotal evidence

suggests that only a small share of the donated items is sold at the stores, much is sold in developing countries, and a significant portion ends up being downcycled, incinerated, or landfilled (e.g., Cobbing et al., 2022). This could reflect the mix of “sales” and premature waste in our model.⁹ Another interpretation, however, is that many donations are no longer socially useful, i.e., they constitute unavoidable waste.

If socially responsible consumers are more numerous (Part A.II), their demand for used products pushes the used price p_u^* above zero. As in the previous subsection, the prospect of a higher resale price raises the total consumption of selfish types relative to that without the secondary market. This means that the effect of the secondary market is in general ambiguous.

In particular, Part B establishes that the secondary market can be doubly harmful again, both increasing production and lowering total private consumption utility. To show this, we construct a situation in which the secondary market leads responsible consumers to replace their new purchases with used products, making them privately worse off. To supply these used goods, selfish consumers inefficiently increase their own consumption of new goods. Furthermore, since the goods are quite fragile, satisfying responsible types’ demand for used goods requires a large volume of new purchases by selfish types. As a result, production is higher than without the secondary market.

If there are many responsible consumers (Part A.III), then their demand exceeds used supply, so in equilibrium they have to consume both products. Since this would mean that they are indifferent, a strict equilibrium does not exist. We return to this issue in our general model in Section 4.5.

Recycling interpretation We conclude this subsection by identifying a reinterpretation of our model, and noting the implications. Suppose that instead of new and used goods, consumers can buy new and recycled versions of the same single-use product (e.g., plastic bottles). After use, one unit of either product can be converted at cost l into $1 - f$ units of the recycled product. The consumption utility from new and recycled products is the same; in particular, there is no disutility l from consuming a recycled product. Consumers receive a payment p_u for selling a used item to be recycled, and the unit price of a recycled product is $(p_u + l)/(1 - f)$. In this reinterpretation, $l > (1 - f)P$ means that from a private perspective, recycling is not economical (it is cheaper to produce new goods every period).

⁹ While our model predicts a used price of zero when there is excess supply, thrift stores typically have positive prices. Transaction costs can account for this difference. Suppose, for instance, that donating items requires an effort cost of e per item. Then, an equilibrium analogous to that above features $p_u^* = e$. Similarly, in the context of recycling below, the small payment for returning empty bottles or other goods after use could compensate for transaction costs.

Notice that with a secondary market, this variant of our model generates the same consumer behavior, and hence the same strict equilibrium, as our original model. Indeed, buying $1/(1-f)$ units of the used product in the original model generates the same net utility as buying one unit of a recycled product does in the new model. Even in the absence of a secondary market, selfish types — who only buy new — consume the same amount in the two models. Hence, if there are sufficiently few responsible types (Part A.I of Proposition 2), then the recycling system lowers production and raises welfare.¹⁰ In particular, this case obtains if f is relatively low, so that supply from selfish consumers is sufficient to satisfy responsible demand. We conclude that due to the presence of responsible consumers, a recycling system with sufficiently low loss f is always beneficial.¹¹

3.3 Dumping Unpalatable Goods

Finally, we assume that $l > (1-f)(P + \kappa)$. This means that it is optimal to replace used goods with new even taking social responsibility into account.

Then, both selfish and responsible consumers strictly prefer new goods over used goods for any $p_u \geq 0$. Thus, there cannot be trade on the secondary market, and we get:

Observation 1 (Irrelevant Secondary Market). *In any strict equilibrium, $p_u^* = 0$. Production and welfare are identical to those without the secondary market.*

4 General Model and Results

In this section, we extend our analysis by allowing for a general distribution of social coefficients k , and considering all steady-state competitive equilibria, not just strict ones. We show that the main insights developed above survive, and we use our methods to investigate other economically important questions. Throughout, we assume that the distribution G of social coefficients k has continuous density g other than at atoms, and the smallest interval containing its support is $[\underline{k}, \overline{k}]$.

¹⁰ This is the case even though responsible types' consumption without the secondary market is not the same in the two models. In the absence of recycling, responsible consumers must dispose of their single-use items. Nevertheless, recycling still lowers the new-good consumption of responsible types from a positive level to zero.

¹¹ We have not discussed the recycling interpretation in the previous case of high-value used products. In the recycling interpretation, the high-value condition $l < (1-f)P$ means that recycling is economical even from a private perspective ignoring externalities. Although there are products (e.g., aluminum and steel) for which this may hold, they are intermediate products not directly sold to consumers. Hence, our model does not apply in its current form.

4.1 Definition of Consistency

We first precisely define the notion of consistency introduced in Section 2, still building on Kaufmann et al.’s (2024) framework for competitive equilibria with vanishingly small consumers. Informally, r_n^* and r_u^* are consistent if consumers who take r_n^* and r_u^* as given respond in a way that generates r_n^* and r_u^* . To flesh out what this rational-expectations-equilibrium-like condition means, we take the following steps. (a) We use consumers’ utility functions to determine per-period demand given r_n^* and r_u^* . (b) We model the effects of an individual’s consumption — which are vanishingly small relative to the market — as marginal shifts in the preceding demand that marginally shift market-balancing production. In Appendix D, we present and modify part of Kaufmann et al.’s analysis to motivate such a definition of a consumer’s impact. In our model, the following effects obtain. Consuming a new good has a direct, one-to-one effect on current production, while consuming a used good can have an indirect effect by raising the used price p_u and thereby inducing others to buy new. Further, any current production caused by either consumption can have indirect effects on future production by raising the future supply of used goods and thus lowering future used prices. (c) We sum these market-balancing production effects of a small individual’s current new and used consumption, and require that they equal r_n^* and r_u^* , respectively.

Formally, we say that r_n^* and r_u^* are consistent if all of the following hold. (a) The consumer’s objective $U_k(c_n, c_u)$ defined in (1) has a maximum for each $k \in [\underline{k}, \bar{k}]$. This implies that in combination with G , consumer optimization generates a per-period average gross demand curve $(D_n(p_u), D_u(p_u)) = E_G[\arg \max_{c_n, c_u} U_k(c_n, c_u)]$, which in general is a correspondence. (b) For a shift in the *current* curve to $(\Delta_n + D_n(p_u), \Delta_u + D_u(p_u))$, the current market-balancing p_u and production are unique in a neighborhood of the equilibrium values $\Delta_n = \Delta_u = 0$ and steady-state level of gross used supply, and at the equilibrium values they are differentiable in Δ_n and Δ_u .¹² Any change in production also changes future used supply, whose effect equals minus the effect of a shock to used demand. Recursively, therefore, differentiability of the contemporaneous effects implies that production in each future period is also differentiable in Δ_n and Δ_u . (c) The intertemporal sums of the derivatives of production with respect to Δ_n and Δ_u exist, and equal r_n^* and r_u^* , respectively.

¹² The above uniqueness and differentiability hold whenever the market-balancing p_u and production are fully determined by market forces, i.e., when either $p_u^* > 0$, or $p_u^* = 0$ and almost all consumers have a strict preference between the products. Below, we briefly consider what happens with an alternative requirement for the case of $p_u^* = 0$ and indifference.

4.2 Harmful Secondary Markets

We now analyze our model when $l = 0$. Our methodology applies equally well, but the analysis is notationally more cumbersome, if $0 < l < (1 - f)P$.

To proceed, we reformulate the equilibrium conditions in terms of the contemporaneous cross-market effect of used consumption on production, defined as the derivative of *current* production with respect to Δ_u , and denoted by Q_c^* . Notice that Q_c^* is also the negative of the derivative of current production with respect to current used supply. We explain here how to write r_n^* and r_u^* as a function of Q_c^* ; then, since r_n^* and r_u^* determine demand, and Q_c^* depends on demand, consistency boils down to an implicit condition on Q_c^* . Indeed, let us trace out the effect of increasing today's production by one marginal unit on total production over time. This quantity is r_n^* ; and since raising current used consumption by a unit raises current production by Q_c^* units and affects future production only through this channel, we have $r_u^* = Q_c^* r_n^*$. Now if current production rises by one unit, used supply next period rises by one unit. Hence, production next period decreases by Q_c^* units. Combining these effects, used supply in two periods rises by $(1 - f) - Q_c^*$ units. This lowers production in two periods by $Q_c^*((1 - f) - Q_c^*)$ units. Continuing this logic, the total impact of raising current production is

$$r_n^* = 1 - Q_c^* - Q_c^*(1 - f - Q_c^*) - Q_c^*(1 - f - Q_c^*)^2 - \dots = 1 - \frac{Q_c^*}{1 - (1 - f - Q_c^*)} = \frac{f}{f + Q_c^*}. \quad (4)$$

Our main result in this subsection is:

Proposition 3 (Secondary Markets are Weakly Harmful).

I. There is a steady-state competitive equilibrium in which $p_u^ = (1 - f)P$, $r_u^* = (1 - f)r_n^* = (1 - f)f$, and all consumers are indifferent between new and used goods. Each consumer's total consumption and social welfare are the same as without the secondary market.*

II. In any other steady-state competitive equilibrium, $(1 - f)P < p_u^ < P$, $r_u^* < (1 - f)r_n^*$, and almost all consumer type k 's total consumption is strictly higher, while social welfare is strictly lower, than without the secondary market.*

III. The latter type of equilibrium does not exist if $g(k)$ is sufficiently high for all $k \in [k, \bar{k}]$ where it exists. Fixing the other primitives, for any G there is a $\hat{G}$ arbitrarily close in the $L1$ norm such that with k distributed according to $\hat{G}$, the latter type of equilibrium exists.

By Part I, there is always an equilibrium that replicates outcomes without the secondary market.

For an intuition, consider individuals' decision between consuming a unit of the used good and consuming $1 - f$ units of the new good — which yield the same private consumption utility. If these two options also generate the same impact on production ($r_u^* = (1 - f)r_n^*$), then everyone is indifferent between them, so that the secondary market equilibrates at the price $p_u^* = (1 - f)P$ following any small shock Δ_n or Δ_u . As a result, a single individual's marginal effect on prices and hence others' total consumption is zero. Since the supply of used products is fixed, buying a unit displaces a unit of used consumption by others, who — being indifferent between the products and having fixed total consumption — buy $1 - f$ units of the new product instead. Hence, consuming one unit of the used product and consuming $1 - f$ units of the new product must raise production by the same amount ($r_u^* = (1 - f)r_n^*$). This implies both that consistency is satisfied, and that an individual has the same incentives as without a secondary market — where any marginal consumption comes from new items — so she consumes the same total amount.

Part II says that if any other equilibrium occurs, welfare is lower than without the secondary market. Generalizing the logic from Section 3.1, a one-unit used-good purchase has a lower effect on production than a $1 - f$ -unit new-good purchase ($r_u^* < (1 - f)r_n^*$). This fosters used consumption by more responsible types, pushing up the used price p_u^* . At the same time, less responsible types purchase new goods, and are encouraged in doing so by the relatively high level of p_u^* .

Part III states that the second, inferior kind of equilibrium does not exist if the distribution G of social coefficients k is sufficiently dense everywhere, but it always exists for some distributions close to G . To see the logic, consider again the decision between consuming a unit of the used good and consuming $1 - f$ units of the new good, and suppose that the former option is significantly less harmful ($r_u^* < (1 - f)r_n^*$). If G is sufficiently steep everywhere, a small increase in the used price p_u pushes many consumers toward new goods, so the effects of the two consumption options must in fact be similar ($r_u^* \approx (1 - f)r_n^*$). But with the two options having similar impacts on production, consumers are even more responsive to the used price, pushing the effects of the options even closer. In this manner, the equilibrium collapses. Hence, for the inferior equilibrium to exist, $g(k)$ must be sufficiently small at least within a range, for instance due to a bimodal or widely dispersed distribution of k . The range necessary to ensure existence of an inferior equilibrium can be arbitrarily narrow.

The logic behind Proposition 3 has the interesting implication that outcomes may be non-monotonic in the population's overall degree of social responsibility:

Proposition 4. *Fix all primitives other than G and u , and take any $\epsilon > 0$ and $W > 0$. There are*

u , G_l , and G_h such that G_h first-order stochastically dominates G_l , and:

(i) if $G = G_l$, then only the equilibrium in Part I of Proposition 3 exists;

(ii) if $G = G_h$, then there are at least two equilibria, with social welfare being within ϵ of that with $G = G_l$ in the best equilibrium and at least W lower in the worst.

The proposition says that an increase in the population's social coefficients can substantially lower welfare in the worst equilibrium without much affecting welfare in the best equilibrium. To see this, suppose that we start from a single-peaked distribution for which only the good equilibrium exists, and consider a slight increase in a small share's social coefficients that makes the distribution bimodal. As a result, an equilibrium may arise in which used consumption has a small effect on production, and almost all individuals consume more.¹³

4.3 New Products

Both in reaction to and independently of our theory, we have heard many individuals (both economists and non-economists) claim that purchasing *new* durable goods is harmless. The argument is based on the sequential nature of production and sales: typically, a good is sold after its production is complete and its production externalities are therefore sunk, so purchasing it cannot cause harm on the margin. At first sight, this logic resembles that in our harmful equilibrium (Proposition 1 and Proposition 3, Part II), where purchasing a good produced in the past is less harmful than purchasing a new good. Nevertheless, we show that from the perspective of our model, the argument is incorrect.

To investigate the issue, we continue to impose $l = 0$, and assume that our “secondary” market arises almost immediately after production. Formally, we let the length of a period become very short, so that the depreciation f is very small. To keep the total utility from consumption on the same scale, we impose that $u(c) = fv(c)$, where $v(\cdot)$ is a baseline utility function that satisfies the conditions on $u(\cdot)$ we have imposed previously. Recall that without a secondary market, a consumer chooses $u'(c_k^{-sm}) = f(P + k)$, which now reduces to $v'(c_k^{-sm}) = P + k$. Hence, as is natural when only new products are available for purchase, baseline consumption is independent of the period length, i.e., our scaling makes sense.

Proposition 5 (New Products with Sunk Externality). *Suppose there is an (arbitrarily small) mass of selfish consumers ($\underline{k} = 0$ and $G(0) > 0$). For a sufficiently small f , only Proposition 3's*

¹³ Although using discrete distributions is simpler, our proof establishes the proposition for distributions that have no gaps in their supports. We do so to verify that the argument is robust.

type-I equilibrium exists.

The proposition implies that if there are some selfish consumers, then equilibrium outcomes with the quick resale of new products are the same as when consumers purchase made-to-order products from manufacturers directly. Intuitively, there are some selfish consumers who are willing to purchase from the manufacturer and resell to others. Since the period length is short, consumers purchase only a little bit per period, so a few such selfish consumers are sufficient to serve the market. Crucially, responsible consumers know that if they buy more, selfish consumers restock more, so the purchase has the same externality impact as buying from the manufacturer.

In our model, rational consumers double as retailers, holding and consuming the product for a short period before selling it to others. This allows us to compare markets for essentially new products with markets for substantively different used products in the same framework. In reality, however, new products are often sold by retailers who specialize in this business and do not consume their inventory while holding it. Incorporating a fully fledged retail sector with inventory management into our theory is beyond the scope of the paper; this would require combining an existing framework with consumers who care about how their purchases affect inventories. Nevertheless, straightforward application of existing work yields a clear conclusion. First, the typical assumption in the literature is that fluctuations in demand are absorbed through changes in inventory rather than adjustments in the retail price (e.g., Blinder and Maccini, 1991). Second, canonical models starting from Scarf (1960) imply that if a consumer purchases extra items, the retailer's future orders increase one-to-one. This again indicates that the externality impact of buying a retail product is identical to that of buying a made-to-order new product.¹⁴

4.4 Beneficial Secondary Markets

We now assume that $l > (1 - f)P$ — i.e., used goods are privately unpalatable — and show that the logic of the potential beneficial equilibrium with secondary markets applies unchanged. To do so, we look for an equilibrium in which $p_u^* = 0$ and there is an excess supply of used goods, yielding $r_u^* = 0$ and $r_n^* = 1$. Then, individuals with k satisfying $(1 - f)(P + k) < l$ prefer new goods, consuming

¹⁴ A simple way to combine an inventory model with ours is the following. Retailers are firms who have a technology to keep items new for free. They order from producers at the end of each period, with the order delivered at the beginning of next period. Consumers can only purchase from retailers. While aggregate demand is deterministic in equilibrium, retailers face bounded idiosyncratic demand uncertainty (due to uncertainty in who consumers purchase from). Each retailer must serve all demand. In this situation, each retailer stocks up to the maximum possible demand. Consumers can purchase extra items from retailers selling less than that. Doing so leads the retailer to restock one-to-one.

an amount c_k^* satisfying $u'(c_k^*) = P + k$ in each period. In contrast, individuals with k satisfying $(1 - f)(P + k) > l$ prefer used goods, consuming an amount c_k^* satisfying $(1 - f)u'((1 - f)c_k^*) = l$ in each period. Using this to express that supply exceeds demand gives:

Proposition 6. *There is a steady-state competitive equilibrium with $p_u^* = 0$, $r_u^* = 0$, and $r_n^* = 1$ if and only if*

$$\int_{\underline{k}}^{l/(1-f)-P} u'^{-1}(P + k) dG(k) > \frac{f}{1-f} \cdot \left(1 - G(l/(1-f) - P)\right) u'^{-1}(l/(1-f))$$

and G has no atom at $k = l/(1-f) - P$. In this equilibrium, production is weakly lower, and social welfare is weakly higher, than without the secondary market. If $G(l/(1-f) - P) < 1$, then production is strictly lower, and social welfare is strictly higher, than without the secondary market.

The result essentially generalizes Proposition 2, Part I, the case in which the secondary market raises welfare by reducing premature waste. The only exception is the non-generic situation in which the distribution of social coefficients k has an atom at exactly $k = l/(1-f) - P$, leading to a non-existence issue. Recall from Section 4.1 that equilibrium requires the market's reaction to a consumer's purchases to be uniquely determined by preferences and market balance. In the candidate equilibrium in the proposition, consumers with $k = l/(1-f) - P$ are indifferent between new and used products. If there is a positive mass of them, then due to an excess supply of used products, the market's reaction to a shock is not unique. We take up this issue at the end of the next subsection.

4.5 Equilibrium Non-Existence and a Resolution

We briefly return to the setting of Section 3.2: we assume that shares g and $1 - g$ of consumers have $k = 0$ and $k = \kappa$, respectively, and $(1 - f)P < l < (1 - f)(P + \kappa)$. We do so to identify and discuss a potential generic non-existence of steady-state competitive equilibrium, and how the non-existence can be resolved.

Proposition 7. *Let $\underline{g}$ be as defined in Proposition 2. If $g < \underline{g}$, then a steady-state competitive equilibrium does not exist.*

If the share of socially responsible consumers is high, then — extending Proposition 2's result on strict equilibria — there is no equilibrium of any kind. Since responsible consumers have demand that exceeds used supply, in any equilibrium some of them must buy new goods, while others buy

used. This would, however, create a free-riding situation in which responsible consumers strictly prefer to buy new and let others buy used, contradicting equilibrium. More precisely, if responsible consumers bought both new and used goods, they would have to be indifferent between a unit of the used good and $1 - f$ units of the new good. Then, due to the cross-market effect, new production would not depend on which option a person chooses: if she buys one unit of the used good, she depletes used supply, so that others buy $1 - f$ more units of the new good. Hence, for any $p_u \geq 0$ the person strictly prefers new goods.

Equilibrium does, however, exist under a minor relaxation of our requirements. Similarly to the situation at the end of Section 4.4, the market's reaction to a purchase is not uniquely determined when $p_u^* = 0$ and responsible consumers are indifferent between the two products, ruling out such equilibria. But it is natural to allow other ways of tying down trade in this specific case. As an example, we can introduce the equilibrium object $\rho \in [0, 1]$, which equals the derivative of aggregate used consumption with respect to used supply, and the negative of the derivative of others' used consumption with respect to an individual's used consumption.¹⁵ Such effects can arise if a consumer looks for used goods through (costless) search, and her success rate depends on available supply and others' search intensities. Clearly, $\rho < 1$ must in equilibrium mean that some products are discarded; otherwise, the equilibrium effect of an increase in an individual's used consumption, namely an increase of $1 - \rho$ in aggregate consumption, would not be feasible.

In the above model, an equilibrium exists for any $g < \underline{g}$, and satisfies $p_u^* = 0$ and $(1 - f)(P + \kappa) = \kappa(1 - f)\rho^* + l$.¹⁶ Intuitively, the free-riding problem above induces responsible types to lower used consumption and thereby reduce the used price to zero as well as waste some used products ($\rho^* < 1$). The resulting market slack makes responsible types indifferent between the products, ensuring that at least some used goods are consumed.

It is easy to check that in this equilibrium, the total consumption of both selfish and responsible types is the same as without the secondary market. Hence, production is higher if and only if responsible types' used consumption is lower, which our specification so far does not tie down. One first-pass additional assumption is that ρ also equals the fraction of used goods from before that is

¹⁵As will be clear from the next paragraph and footnote, ρ is closely related to the cross-market effect we have worked with so far (consumers treat ρ as they would $(1 - f)^{-1}Q_c$, only the consistency condition differs). We use a consumer's effect on others' used consumption because it is more natural for the arguments that follow in this paragraph.

¹⁶The latter imposes that a responsible consumer is indifferent between consuming $1 - f$ units of the new good and 1 unit of the used good. The current disutility from $1 - f$ units of the new good is $(1 - f)(P + \kappa)$, and that from a used good is $\kappa(1 - f)\rho^* + l$, where the first term comes from the cross-market effect $(1 - f)\rho^*$ of used consumption on others' new consumption. Either choice increases future used gross supply by $1 - f$ units, so their impacts on the future are identical.

consumed by the group of indifferent consumers, with the rest discarded. (In general, this second, average effect of an increase in gross used supply could be different from the previous, marginal effect.) Then, since $\rho^* = (P + \kappa - l/(1 - f))/\kappa$ can be arbitrarily low, the level of production can be higher than without the secondary market. Indeed, if most consumers are responsible, then — perversely — the secondary market is always harmful. This is because responsible consumers do not generate premature waste without the secondary market, but do generate some with it. The harmful effect now occurs because the secondary market encourages free-riding. This example also demonstrates that an equilibrium used price of zero does not guarantee that the secondary market raises welfare.

The same relaxation of equilibrium requirements also resolves the equilibrium non-existence issue in Proposition 6. In that case, we can posit that indifferent consumers purchase new goods for all of their consumption, yielding $\rho^* = 0$. Then, the stated equilibrium exists for all G satisfying the inequality in the proposition.

5 Conclusion

Our analysis suggests several questions for future research that can be studied with modifications of the techniques we have developed. Most importantly, while we have considered the welfare effects of secondary markets, we have not analyzed policy interventions. It would be natural to study how primary-market policies, such as taxes, caps, or carbon tariffs, interact with the effects we have found, and to explore potential policies that are specific to the secondary market. As an example, some public or non-profit organizations distributing used goods may make a policy of charging no or minimal prices, and rationing if there is excess demand. On the one hand, this donation-type system may improve outcomes by ruling out one main welfare-decreasing mechanism we have found, that a high used price encourages new purchases. On the other hand, it can lead to offsetting effects of both old and new kinds. A well-known concern is that the availability of free goods encourages recipients to consume more. A new concern is that a “resale-based justification effect” may encourage responsible donors to consume more new goods. Such an effect arises because responsible consumers know that by donating used products, they lower new purchases by others — which justifies higher consumption of new products exactly because they are socially responsible. A parent may, for instance, buy more clothes for her baby because she knows that donating the clothes will prevent new purchases later.

There are also some ingredients that are natural and straightforward to add to our framework. For instance, while we have taken the fragility f of products to be exogenous, it is fruitful to ask how this will be determined when the choice is endogenous. As an immediate point, socially responsible consumers dislike the harm they cause by replacing broken items, so they tend to prefer more durable products. But durability choice when consumers with different social coefficients interact appears to be a non-trivial equilibrium phenomenon.

Finally, there are relevant questions that call for more fundamental modifications of our theory. Perhaps most importantly, our framework ignores socially responsible consumers' potential motive to alter the beliefs of firms, investors, innovators, and policymakers. In general, a consumer may buy expensive green products to signal that bringing these to the market is worth it. And in the context of secondary markets, a consumer may be reluctant to raise the price of used goods, lest she encourage future purchases of new goods. This can motivate responsible consumers to buy fewer or supply more used goods. To account for such motives, it is necessary to integrate uncertainty and inferences into our framework.

References

- Aghion, Philippe, Roland Bénabou, Ralf Martin, and Alexandra Roulet**, “Environmental Preferences and Technological Choices: Is Market Competition Clean or Dirty?,” *American Economic Review: Insights*, 2023, 5 (1), 1–20.
- Anderson, Simon P. and Victor A. Ginsburgh**, “Price Discrimination Via Second-Hand Markets,” *European Economic Review*, 1994, 38 (1), 23–44.
- Andre, Peter, Teodora Boneva, Felix Chopra, and Armin Falk**, “Misperceived Social Norms and Willingness to Act Against Climate Change,” *Review of Economics and Statistics*, forthcoming.
- Arnold, Lutz G.**, “On the Neutrality of Socially Responsible Investing,” *Theoretical Economics*, 2023, 18 (1), 65–95.
- Blinder, Alan S. and Louis J. Maccini**, “Taking Stock: A Critical Assessment of Recent Research on Inventories,” *Journal of Economic Perspectives*, 1991, 5 (1), 73–96.
- Borusiak, Barbara, Andrzej Szymkowiak, Elena Horska, Natalia Raszka, and Elżbieta Żelichowska**, “Towards Building Sustainable Consumption: A Study of Second-Hand Buying Intentions,” *Sustainability*, 2020, 12 (3), 875.
- Broccardo, Eleonora, Oliver Hart, and Luigi Zingales**, “Exit versus Voice,” *Journal of Political Economy*, 2022, 130 (12), 3101–3145.
- Bulow, Jeremy I.**, “Durable-Goods Monopolists,” *Journal of Political Economy*, 1982, 90 (2), 314–332.

- Coase, Ronald H.**, “Durability and Monopoly,” *Journal of Law and Economics*, 1972, 15 (1), 143–149.
- Cobbing, Madeleine, Sodfa Daaji, Mirjam Kopp, and Viola Wohlgemuth**, “Poisoned Gifts: From Donations to the Dumpsite: Textiles Waste Disguised as Second-Hand Clothes Exported to East Africa,” Technical Report, Greenpeace 2022.
- Dewatripont, Mathias and Jean Tirole**, “The Morality of Markets,” *Journal of Political Economy*, 2024, 132 (8), 2655–2694.
- Dufwenberg, Martin, Paul Heidhues, Georg Kirchsteiger, Frank Riedel, and Joel Sobel**, “Other-Regarding Preferences in General Equilibrium,” *Review of Economic Studies*, 2011, 78 (2), 613–639.
- Edbring, Emma Gullstrand, Matthias Lehner, and Oksana Mont**, “Exploring Consumer Attitudes to Alternative Models of Consumption: Motivations and Barriers,” *Journal of Cleaner Production*, 2016, 123, 5–15.
- Gatzer, Sebastian, Stefan Helmcke, and Daniel Roos**, “Playing Offense on Circularity Can Net European Consumer Goods Companies €500 Billion,” Technical Report, McKinsey & Company 2022.
- Green, Daniel and Benjamin Roth**, “The Allocation of Socially Responsible Capital,” *Available at SSRN 3737772*, 2021.
- Guiot, Denis and Dominique Roux**, “A Second-hand Shoppers’ Motivation Scale: Antecedents, Consequences, and Implications for Retailers,” *Journal of Retailing*, 2010, 86 (4), 355–371.
- Hakenes, Hendrik and Eva Schliephake**, “Responsible Investment and Responsible Consumption,” *Management Science*, forthcoming.
- Hamari, Juho, Mimmi Sjöklint, and Antti Ukkonen**, “The Sharing Economy: Why People Participate in Collaborative Consumption,” *Journal of the Association for Information Science and Technology*, 2016, 67 (9), 2047–2059.
- Hendel, Igal and Alessandro Lizzeri**, “Interfering with Secondary Markets,” *RAND Journal of Economics*, 1999, 30 (1), 1–21.
- and —, “The Role of Leasing under Adverse Selection,” *Journal of Political Economy*, 2002, 110 (1), 113–143.
- Herweg, Fabian and Klaus M. Schmidt**, “How to Regulate Carbon Emissions with Climate-Conscious Consumers,” *Economic Journal*, 2022, 132 (648), 2992–3019.
- Kaufmann, Marc, Peter Andre, and Botond Köszegi**, “Understanding Markets with Socially Responsible Consumers,” *Quarterly Journal of Economics*, 2024, 139 (3), 1989–2035.
- Klooster, Astrid, Blanca C. Bellostas, Marvin Henry, and Li Shen**, “Do We Save the Environment by Buying Second-Hand Clothes? The Environmental Impacts of Second-Hand Textile Fashion and the Influence of Consumer Choices,” *Journal of Circular Economy*, 2024.
- Krahnen, Jan, Jörg Rocholl, and Marcel Thum**, “A Primer on Green Finance: From Wishful Thinking to Marginal Impact,” *Review of Economics*, 2023, 74 (1), 1–19.

- Meier, Jean-Marie, Henri Servaes, Jiaying Wei, and Steven Chong Xiao**, “Do Consumers Care About ESG? Evidence from Barcode-Level Sales Data,” 2023. Working Paper.
- Moisson, Paul-Henri**, “Ethics and Impact Investment,” 2020. Working Paper.
- Norwood, F. Bailey and Jayson L. Lusk**, *Compassion, by the Pound: The Economics of Farm Animal Welfare*, Oxford University Press, 2011.
- Oehmke, Martin and Marcus M. Opp**, “A Theory of Socially Responsible Investment,” *Review of Economic Studies*, 2025, 92 (2), 1193–1225.
- Pástor, Ľuboš, Robert F. Stambaugh, and Lucian A. Taylor**, “Sustainable Investing in Equilibrium,” *Journal of Financial Economics*, 2021, 142 (2), 550–571.
- Piccolo, Alessio, Jan Schneemeier, and Michele Bisceglia**, “Externalities of Responsible Investments,” *Working Paper*, 2022.
- Rodemeier, Matthias**, “Willingness to Pay for Carbon Mitigation: Field Evidence from the Market for Carbon Offsets,” 2023. Working Paper.
- Rodrigues, Maria, João F. Proença, and Rita Macedo**, “Determinants of the Purchase of Secondhand Products: An Approach by the Theory of Planned Behaviour,” *Sustainability*, 2023, 15 (14).
- Rust, John**, “When is it Optimal to Kill off the Market for Used Durable Goods?,” *Econometrica*, 1986, 54 (1), 65–86.
- Sandin, Gustav and Greg M. Peters**, “Environmental Impact of Textile Reuse and Recycling – A Review,” *Journal of Cleaner Production*, 2018, 184, 353–365.
- Scarf, Herbert**, “The Optimality of (S, s) Policies in the Dynamic Inventory Problem,” in Kenneth J. Arrow, Samuel Karlin, and Patrick Suppes, eds., *Mathematical Methods in the Social Sciences*, Stanford University Press, 1960.
- Schulze Tilling, Anna**, “Changing Consumption Behavior with Carbon Labels: Causal Evidence on Behavioral Channels and Effectiveness,” 2024. Working Paper.
- Sieper, Edward and Peter L. Swan**, “Monopoly and Competition in the Market for Durable Goods,” *Review of Economic Studies*, 1973, 40 (3), 333–351.
- Sobel, Joel**, “Do Markets Make People Selfish,” *Working Paper*, 2007.
- Styvén, Maria Ek and Marcello M. Mariani**, “Understanding the Intention to Buy Second-hand Clothing on Sharing Economy Platforms: The Influence of Sustainability, Distance from the Consumption System, and Economic Motivations,” *Psychology & Marketing*, 2020, 37 (5), 623–753.
- Swan, Peter L.**, “Durability of Consumption Goods,” *American Economic Review*, 1970, 60 (5), 884–894.
- ThreadUp**, “Resale Report 2024,” Technical Report, Based on GlobalData 2023 Market Sizing and Growth Estimates 2024.
- Trammell, Philip**, “Ethical Consumerism,” 2023. Working Paper.

Turunen, Linda Lisa Maria and Hanna Leipämaa-Leskinen, “Pre-Loved Luxury: Identifying the Meanings of Second-hand Luxury Possessions,” *Journal of Product & Brand Management*, 2015, 24 (1), 57–65.

Varah, Franky, Mirinchonme Mahongnao, Balaram Pani, and Sophayo Khamrang, “Exploring Young Consumers’ Intention toward Green Products: Applying an Extended Theory of Planned Behavior,” *Environment, Development and Sustainability*, 2021, 23, 9181–9195.

Waldman, Michael, “Durable Goods Pricing When Quality Matters,” *Journal of Business*, 1996, 69 (4), 489–510.

A Naive and Deontological Consumers

Our model posits rational consequentialist consumers, who correctly understand and care about their net effect on the externality. We now briefly discuss consumers who are naive in the sense of Kaufmann et al. (2024). A naive consumer understands or thinks about only her direct effect, so in our setting she believes that buying a new good causes an externality, while buying a used good does not. Hence, a naive consumer chooses $c_n \geq 0$ and $c_u \geq 0$ to maximize $u(c_n + (1 - f)c_u) - lc_u - Pc_n - p_uc_u + p_u^*(c_n + (1 - f)c_u) - kc_n$. Since naive consumers have standard preferences defined over their own consumption, a classical definition of competitive equilibrium applies, and standard analytical tools can be used. Note that the behavior of naive consumers coincides with that of deontological consumers, who dislike buying new products out of principle in proportion to their social coefficients k . Hence, our analysis applies unchanged to deontological consumers as well.

With no secondary market, naive consumers have the same belief about new products as rational consumers, so they consume the same amount. Furthermore, in any strict equilibrium with a rational population, responsible consumers evaluate their purchases in terms of direct effects, so the same outcome is also an equilibrium with naive consumers. Hence, in a strict equilibrium, the welfare effects of a secondary market do not depend on whether consumers are rational or naive.

Finally, we reconsider the general model from Section 4.2 (where $l = 0$) with naive consumers. We obtain:

Proposition 8 (Secondary Markets with Naive Consumers). *There is a unique steady-state competitive equilibrium. In this equilibrium, welfare is strictly lower, and almost all consumer type k ’s total consumption is strictly higher, than without the secondary market.*

With naive consumers, the secondary market always strictly lowers welfare. To see this, we can distinguish two cases. If $p_u^* = (1 - f)P$, then naive consumers think that by buying used, they can

avoid generating an externality at no cost to themselves. Hence, all naive consumers with $k > 0$ strictly prefer used goods, and have higher total consumption than without the secondary market. This increase in consumption lowers welfare. If $p_u^* > (1 - f)P$, then a situation akin to that in the bad equilibrium with rational consumers arises. Less responsible consumers buy new goods and sell their used goods on the secondary market, and are encouraged in doing so by the high resale price. More responsible consumers in turn buy used, and because they believe they are not causing an externality, they consume a lot too. With everyone consuming more, production is higher and welfare is lower than without the secondary market.

Comparing Propositions 3 and 8, it is immediate that the best equilibrium with rational consumers yields higher welfare than the equilibrium with naive consumers. This is because rational consumers are capable of recognizing the link between the primary and secondary markets. Similarly, comparing Propositions 5 and 8 implies that a market for new goods that have already been produced may yield higher welfare with rational consumers. This is the case if naive consumers believe that buying such goods is harmless. However, we have found that for used products, the welfare comparison between a bad equilibrium with rational consumers and the equilibrium with naive consumers is ambiguous.

We have restricted attention to all-rational and all-naive populations of consumers, but future work can analyze markets with a mix of consumer types. We conjecture that this interaction can introduce further welfare-decreasing effects. Consider, for instance, a population of rational and naive consequentialist consumers with $l = 0$. If most are rational, then the used price p_u must be relatively low, as the many rational types understand the cross-market effect and thus have low demand for used goods. The low p_u , in turn, encourages naive types to consume a lot of used goods, which they think are harmless. In contrast, if most consumers are naive, then their demand for used goods pushes up the used price p_u . This implies that rational consumers are encouraged to sell used, and thus buy new, goods for two reasons. The first is the familiar price effect, that they can obtain a high resale price. The second is another manifestation of the resale-based justification effect mentioned in the conclusion: that by selling used goods, they lower naive consumers' new purchases and thus the externality they generate.¹⁷

¹⁷ One may also analyze other types of consequentialist motives, such as a preference to reduce premature waste rather than new production. We conjecture that such consumers can only raise the value of the secondary market.

B Heterogeneity in Consumption Utility

We discuss some implications of relaxing our assumption that all consumers have the same consumption utility. We start from a baseline case in which all consumers are selfish, and fractions h and $1 - h$ have disutilities from used consumption equal to $l > 0$ and 0 , respectively, where $l < (1 - f)P$. The former consumers could, for example, be high-income consumers who dislike used products, with the latter being low-income consumers who focus on prices. We confirm that the harmful forces we have identified in Section 3 are still operational. Like in that section, we restrict attention to strict equilibria.

First, consider the baseline case with selfish consumers. In any strict equilibrium with a secondary market, $(1 - f)P - l < p_u^* < (1 - f)P$, low-income consumers buy used goods, high-income consumers buy new goods, both types' consumption is strictly higher than without the secondary market, and production is also higher. With no production externalities, this raises social welfare through the classical allocative benefit of the secondary market — channeling used goods to consumers who value them relatively higher. When there are externalities, this allocative benefit is still present, but the externality from higher production lowers or reverses it.

Second, suppose that low-income consumers become socially responsible. This lowers consumption without the secondary market, but since used consumption does not increase production, it leaves consumption with the secondary market unchanged. Hence, the secondary market lowers the value of social responsibility and vice versa.

Third, suppose that high-income consumers become socially responsible with $k = \kappa > l/(1 - f)$.

Observation 2. *In any strict equilibrium, we have $p_u^* > (1 - f)P$, high-income consumers buy used goods, and low-income consumers buy new goods. Production is strictly higher, and social welfare is strictly lower, than without the secondary market.*

In equilibrium, high-income consumers buy used goods to avoid generating an externality, thereby raising used prices and encouraging low-income consumers to buy new. This means that the secondary market raises the total consumption of both types, increases production, and also *worsens* the allocation in the economy. In particular, if $p_u^* \approx (1 - f)P$, then low-income consumers benefit minimally from the secondary market, while high-income consumers are (privately) harmed, despite consumption and production increasing. Furthermore, it is easy to show that the increase in production can be arbitrarily large and larger than with selfish consumers. Indeed, as κ becomes large, responsible consumers' total consumption without the secondary market declines, but their

total consumption with the secondary market remains unchanged.

C Proofs

Throughout the following proofs we will measure the total consumption of consumers and of the population in units of *new consumption*, denoted by $c_k := c_{k,n} + (1 - f)c_{k,u}$. Since a unit of the used good has the same utility — absent the distaste factor l , which we incorporate elsewhere — as $1 - f$ units of the new good, it is rescaled. Denoted by $\hat{c}_u := (1 - f)c_u$ units of rescaled used goods. Now, rewrite the consumers' utility function (1) as $U_k(c_n, \hat{c}_u) = u(c_n + \hat{c}_u) - e_n^k c_n - e_u^k \hat{c}_u$ where $e_n^k = P - p_u^* + r_n^* k$ and $e_u^k = (1 - f)^{-1}(l + p_u) - p_u^* + (1 - f)^{-1}r_u^* k$, which do not depend on consumption choices. Using this rewriting, consider the policy function $C : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ that solves $u'(C(e)) = e$ for any $e > 0$. Given this specification optimal consumer behavior is straightforward (it's like a selfish consumer's decision of non-durable perfect substitutes with different prices) and characterized as follows.

Fact 1. *In any equilibrium with a secondary market (1) if $e_n^k < e_u^k$, then the person strictly prefers new goods, and consumes $c_n = C(e_n^k)$ units of new goods; (2) if $e_n^k > e_u^k$, then she strictly prefers used goods, and consumes $\hat{c}_u = C(e_u^k)$ units of rescaled used goods or $c_u = (1 - f)^{-1}C(e_u^k)$ units of used goods; and (3) if $e_n^k = e_u^k$, then she is indifferent and consumes $C(e_u^k) = C(e_n^k) = c_n + \hat{c}_u$ in some combination.*

The bounds on l and k in the main text ensure the following, which we use implicitly throughout:

Fact 2. *Without the secondary market everyone consumes more than socially optimal. Moreover, consumption of used goods generates more welfare than throwing them away.*

The next supporting lemma compares the total consumption with vs without a secondary market.

Lemma 1. *The total consumption of a consumer who is indifferent in an equilibrium with a secondary market is the same as without a secondary market. The total consumption of a consumer who strictly prefers either used or new goods is weakly higher and always strictly so if they consume used and new goods without the secondary market.*

Proof. Consider the following consumption plan. When buying 1 unit in total consumption ($c_n + \hat{c}_u$), buy a fraction f of new and $1 - f$ unit of rescaled used consumption. The effective price of this

bundle is $fe_n^K + (1-f)e_u^k$, which equals

$$f(P - p_u^* + r_n^*k) + (1-f)((1-f)^{-1}(l + p_u) - p_u^* + (1-f)^{-1}r_u^*k) = fP - p_u^* + p_u + l + k(fr_n^* + r_u^*).$$

Now, in equilibrium $p_u = p_u^*$ and, either $r_n^* = 1$ and $r_u^* = 0$ (in strict equilibria) or $r_n^* = \frac{f}{f+Q_c^*}$ and $r_u^* = \frac{Q_c^*f}{f+Q_c^*}$ (with a type distribution). In both cases, $fr_n^* + r_u^* = f$.

Therefore, the effective price is $f(P + k) + l$, the same as for a unit of consumption without secondary markets for the case when the consumer does not throw away used goods without a secondary market. In that case, a consumer can replicate any decision possible without a secondary market in any equilibrium with a secondary market. Moreover, this consumption bundle is optimal with secondary markets if and only if the consumer is indifferent between used and new goods, i.e., $e_n^k = e_u^k$. If instead $e_n^k > e_u^k$ or $e_u^k > e_n^k$, then the person can obtain a strictly lower effective price and thus consumes strictly more in total.

A consumer that throws used goods without a secondary market, can still only consume new and weakly benefits from $p_u^* \geq 0$ and the weakly lower externality impact. Hence, they consume weakly more with a secondary market.

□

Furthermore, the following lemma relates consumption and production.

Lemma 2. *Suppose that $l < (1-f)P$. If with the secondary market everyone's consumption is weakly higher than without and a positive share's consumption is strictly higher, then production is strictly higher.*

Proof. Without a secondary market, as $l < (1-f)P < (1-f)(P+k)$ for $k \geq 0$, all types prefer to keep used goods. Thus no goods are thrown away and in the steady state consumers exactly replace the fraction f of lost value each period by new products. Hence, production equals f times consumption. In any steady state equilibrium with a secondary market, production has to be at least f times overall consumption: Again, consumed goods lose a proportion of f of their value the next period. To keep overall consumption levels steady (as required by equilibrium), at least that proportion of consumption has to be replaced by new goods. Thus, in the steady state, per period production has to be at least f times steady state overall consumption. If total consumption for all consumers is individually weakly higher and strictly for a positive mass with secondary market, then overall consumption is strictly higher. Finally, new production with a secondary market is

therefore strictly higher than production without a secondary market. $\square$

Proof of Proposition 1. With a secondary market and socially responsible consumers, a candidate strict equilibrium requires $(1 - f)P < p_u^* + l < (1 - f)(P + \kappa)$. Since $r_u^* = 0$ and $r_n^* = 1$, the first condition is equivalent to selfish types strictly preferring new good. The second condition is equivalent to responsible ones strictly prefer used. Finally, the no-arbitrage condition $p_u^* < P$ implies that $(1 - f)P - l < p_u^* < \min \{P, (1 - f)(P + \kappa) - l\}$ is necessary in any strict equilibrium.

Next, since both consumer types prefer a type of good strictly, by Lemma 1 both consume strictly more and by Lemma 2 production is strictly higher. Since no used goods are wasted in either case and production is strictly higher, welfare is strictly lower with a secondary market.

Finally, the strict equilibrium exists if and only if g is such that the market-clearing condition holds. Consumers buying used demand $(1 - f)^{-1}C((1 - f)^{-1}l + \frac{f}{1-f}p_u^*)$ units of used goods and supply $(1 - f)$ times that amount from their last period consumption. Consumers preferring new consume $C(P - p_u^*)$ of new goods and supply that amount in used goods from their last period consumption. Hence, market clearing requires $gC(P - p_u^*) = (1 - g)\frac{f}{1-f}C((1 - f)^{-1}l + \frac{f}{1-f}p_u^*)$ or $\frac{g}{1-g} = \frac{f}{1-f} \frac{C((1-f)^{-1}l + \frac{f}{1-f}p_u^*)}{C(P - p_u^*)}$. Notice that the right hand side of this equation is continuous and strictly decreasing in p_u^* . Now let $\underline{g}$ be such that this equation holds for $p_u^* = \min\{(1 - f)(P + \kappa), P\}$, with the convention that dividing the right hand side by $C(0) := \infty$ implies $\underline{g} = 0$, and $\bar{g}$ such that the equation holds with $p_u^* = (1 - f)P - l$. We have $\underline{g}, \bar{g} \in [0, 1)$. By the intermediate value theorem, for any g in between market clearing will be satisfied for some p_u^* in the admissible range. For any g equal or outside, market clearing would only be satisfied for a price that we already ruled out.

Proof of Proposition 2. A. First, we will identify which good the selfish and the responsible consumers buy. Since in every equilibrium $p_u^* \geq 0$ and $(1 - f)P < l$, selfish ($k = 0$) consumers face effective prices $e_n^k = P - p_u^* \leq P < (1 - f)^{-1}l \leq (1 - f)^{-1}(p_u^* + l) - p_u^* = e_u^k$ and choose new. In any strict equilibrium $r_n^* = 1$ and $r_u^* = 0$, and thus socially responsible ($k = \kappa$) consumers prefer new iff

$$e_n^k = P - p_u^* + \kappa < (1 - f)^{-1}l + (1 - f)^{-1}p_u^* - p_u^* = e_u^k \iff (1 - f)(P + \kappa) - l < p_u^*.$$

Notice that we assumed $(1 - f)(P + \kappa) - l > 0$ and hence socially responsible consumers prefer used when $p_u^* = 0$. If they don't prefer used, then no consumer prefers used and there is an oversupply

which is only consistent with $p_u^* = 0$. Therefore, in any strict equilibrium $p_u^* \in [0, (1-f)(P+\kappa)-l)$, socially responsible consumers strictly prefer used and buy $(1-f)^{-1}C((1-f)^{-1}l + \frac{f}{1-f}p_u^*)$ units while selfish consumers strictly prefer new goods and buy $C(P-p_u^*)$ units.

As in the proof of Proposition 1, market clearing is given by $\frac{g}{1-g} = \frac{f}{1-f} \frac{C((1-f)^{-1}l + \frac{f}{1-f}p_u^*)}{C(P-p_u^*)}$ here the right hand side is continuous and strictly decreasing in p_u^* and the left hand side is strictly increasing in g . Now, let $\bar{g}$ be such that $\frac{\bar{g}}{1-\bar{g}} = \frac{f}{1-f} \frac{C((1-f)^{-1}l)}{C(P)}$ and $\underline{g}$ be such that $\frac{\underline{g}}{1-\underline{g}} = \frac{f}{1-f} \frac{C((1-f)^{-1}l + \frac{f}{1-f}p_u^+)}{C(P-p_u^+)}$, where $p_u^+ = (1-f)(P+\kappa)-l$.

I. If $g \in (\bar{g}, 1)$, then at $p_u^* = 0$ the market clearing condition gives $\frac{g}{1-g} > \frac{\bar{g}}{1-\bar{g}} = \frac{f}{1-f} \frac{C((1-f)^{-1}l)}{C(P)}$. Hence there is oversupply, so that $p_u^* = 0$ is an equilibrium. It is also clear that every $p_u^* > 0$ leads to oversupply, and thus can't be part of an equilibrium. Selfish consumers consume the same amount as without a secondary market, so their contribution to production is the same. Socially responsible consumers consume some new goods without a secondary market, whereas with a secondary market they consume none. Hence, their contribution to production is strictly lower. Moreover, since they strictly prefer used their total consumption is higher than without a secondary market by Lemma 1.

Overall, individual consumption levels are weakly higher and production strictly lower, resulting in strictly higher welfare.

II. When $g \in (g, \bar{g})$, then $\frac{g}{1-g} \in (\frac{g}{1-g}, \frac{\bar{g}}{1-\bar{g}})$. By the intermediate value theorem, there exists a unique $p \in (0, p_u^+)$ s.t. $D(p_u^*) = S(p_u^*)$ and thus market clearing holds. Since both consumers strictly prefer one type of good, both consumer types consume strictly more than without a secondary market by Lemma 1.

III. If $g < \underline{g}$, then $\frac{g}{1-g} < \frac{\underline{g}}{1-\underline{g}} < \frac{f}{1-f} \frac{C((1-f)^{-1}l + \frac{f}{1-f}p_u^*)}{C(P-p_u^*)}$ for any admissible p_u^* . Hence, for any possible price, there would be an undersupply of used goods, which is impossible.

B. Take any $p_u^* \in (0, (1-f)(P+\kappa)-l)$. We show that a $u(\cdot)$ exists such that the resulting equilibrium satisfies the statement and has used price p_u^* . We use the following fact:

Fact 3. Suppose $0 < c_1 < c_2 < c_3 < c_4$ and $e_1 > e_2 > e_3 > e_4 > 0$. Then, there is a $u(\cdot)$ satisfying our assumptions such that $C(e_m) = c_m$ for each $m = 1, 2, 3, 4$.

Consider any $M > 0$. By the above fact, for sufficiently small $\epsilon > 0$ we can choose a $u(\cdot)$ such that $C(f(P+\kappa)+l) = M$, $C(\frac{f}{1-f}p_u^* + (1-f)^{-1}l) = M + \epsilon$, $C(P) = M + 2\epsilon$, and $C(P-p_u^*) = M'$, where $M' \equiv (1-g)f(M+\epsilon)/(g(1-f)) > (M+\epsilon)/f > M/f$.

By construction, we are in Case II of Part A, so there is a unique equilibrium. Again by

construction, in the equilibrium the used price is p_u^* , responsible types consume $M + \epsilon$, and selfish types consume M' .

Production per period is gM' with the secondary market, and

$$g(M + 2\epsilon) + (1 - g)fM < g(M + 2\epsilon) + (1 - g)f(M + \epsilon) = g(M + 2\epsilon) + g(1 - f)M'$$

without the secondary market. Since $fM' > M$, for a sufficiently small ϵ production is higher with the secondary market.

Total net private consumption utility with the secondary market is

$$g(u(M') - PM') + (1 - g)(u(M + \epsilon) - l(M + \epsilon)),$$

and without the secondary market it is

$$g(u(M + 2\epsilon) - P(M + 2\epsilon)) + (1 - g)(u(M) - (1 - f)lM - fPM)$$

For a sufficiently small ϵ , both components are larger than above: the first one because $u(M') - u(M + 2\epsilon) < u'(M + 2\epsilon)(M' - (M + 2\epsilon)) = P(M' - (M + 2\epsilon))$, and the second one because $l > P$ and $u'(M)$ is finite. $\square$

Proof of Observation 1. Notice that the effective price of a new good is at most $P + \kappa$. Moreover, the lowest effective price for a used good is $(1 - f)^{-1}l$. When $l > (1 - f)(P + \kappa)$, every consumer will prefer new goods to used goods. Hence, in any strict equilibrium used goods have to be thrown which is only possible when $p_u^* = 0$. Moreover, $l > (1 - f)(P + \kappa)$ also implies that used goods will be thrown without a secondary market. Hence, production, consumption, and welfare are identical across both settings. $\square$

Preparation for Proposition 3 We now turn to Proposition 3.

First, since G has a continuous density except at atoms, it can be decomposed as $G = G_1 + G_2$, where G_1 is absolutely continuous with positive density g , and G_2 consists of atoms, the set of which we denote by $\mathcal{A}(G)$. The mass at atom $k \in \mathcal{A}(G)$ is $\Delta G(k) = G(k) - G(k^-)$. For any measurable function f and a measurable set C , $\int_{k \in C} f(k) dG(k) = \int_{k \in C} f(k)g(k) dk + \sum_{k \in \mathcal{A}(G) \cap C} f(k) \Delta G(k)$.

Recall that we assume $l = 0$ for this part. The proof proceeds as described on page 17 by

expressing the equilibrium conditions in terms of the cross market effect Q_c^* , where $r_n^* = \frac{f}{f+Q_c^*}$ and $r_u^* = Q_c^* r_n^*$. We now consider when Q_c^* is consistent in equilibrium using the demand and supply in equilibrium as determined in fact 1.

First, consider the case $Q_c^* = 1 - f$, so that $(1 - f)r_n^* = r_u^*$. Then the impact on the externality per unit of equivalent consumption is the same, and hence all consumers only care about the (monetary) price of goods (applying Fact 1). Hence the market-balancing price p_u^* must satisfy $p_u^* = (1 - f)P$; otherwise all consumers would strictly prefer one of the two products, violating market clearing. Now, all consumers are indifferent between used and new products. Thus, raising current used consumption by a unit raises current new consumption and therefore production by $(1 - f)$ units. This is consistent iff $Q_c^* = (1 - f)$.

Next consider $Q_c^* < 1 - f$.¹⁸ Let k' be the unique consumer who is indifferent between new and used products ($e_n^{k'} = e_u^{k'}$):

$$k' = (p_u - (1 - f)P) \frac{(f + Q_c^*)}{f((1 - f) - Q_c^*)}.$$

Assume for now that $k' \notin \mathcal{A}(G)$. By fact 1, types $k < k'$ buy new and types $k > k'$ buy used. As a result,

$$D_n(p_u) = \int_{\underline{k}}^{k'} c_k dG(k) \quad \Rightarrow \quad D'_n(p_u^*) = \frac{(f + Q_c^*)}{f((1 - f) - Q_c^*)} g(k^*) c_{k^*},$$

where k^* is the equilibrium indifferent type. Similarly, noting that $\partial c_k / \partial p_u = 1/u''(c_k)$,

$$\begin{aligned} D_n(p_u) &= \int_{k'}^{\bar{k}} c_u(k) dG(k) = \int_{k'}^{\bar{k}} (1 - f)^{-1} c_k dG(k) \quad \Rightarrow \\ D'_u(p_u^*) &= (1 - f)^{-1} \left[-\frac{(f + Q_c^*) g(k^*) c_{k^*}}{f((1 - f) - Q_c^*)} + \int_{k^*}^{\bar{k}} \frac{1}{u''(c_k)} dG(k) \right]. \end{aligned}$$

By market clearing for used goods, we must have $k^* \in (\underline{k}, \bar{k})$, so $D'_u(p_u^*) < 0$. Since the current supply of used products is fixed, we also have that $\Delta_u + D_u(p_u)$ is constant, so that $\partial p_u / \partial \Delta_u = -1/D'_u(p_u^*)$, yielding $Q_c^* = \partial D_n(p_u) / \partial \Delta_u|_{p_u=p_u^*} = D'_n(p_u^*) / (-D'_u(p_u^*))$. Plugging in the above expressions, we conclude that

$$Q_c^* = (1 - f) \left[\frac{(f + Q_c^*) g(k^*) c_{k^*}}{(f + Q_c^*) g(k^*) c_{k^*} - f((1 - f) - Q_c^*) \int_{k^*}^{\bar{k}} (1/u''(c_k)) dG(k)} \right], \quad (5)$$

is the condition for $Q_c \in (0, (1 - f))$ to be consistent. Notice that $Q_c = (1 - f)$ always satisfies

¹⁸The case $Q_c^* > 1 - f$, is analogous, but the resulting consistence condition will be such that $Q_c^* < (1 - f)$ after all which would be a contradiction.

condition (5).

Assume now that $k' \in \mathcal{A}(G)$. Then a mass of consumers is indifferent. If the consumption of this mass is split between used and new goods, then for any small shock the consumption has to remain split, since otherwise market clearing would fail. Hence, there can't be a price change in response to a shock, which means consumers consumption levels don't react to a shock and thus an increase of a unit of used consumption ($(1 - f)$ in rescaled units) leads to an increase of $(1 - f)$ units of production. That is, $Q_c^* = 1 - f$. If the consumption of the mass of indifferent consumers is not split, then the market reaction to a shock is not differentiable and thus no equilibrium exists: If all indifferent consumers consume used goods and the shock to used demand is negative (demand for used goods falls), then market-clearing implies that the price for used goods falls and Q_c would be calculated as above. However, if the shock is positive (demand increases) then the price cannot rise as this would shift a mass of consumption towards new goods. Instead the price would remain the same yielding $Q_c = (1 - f)$.¹⁹

Note that with $Q_c = (1 - f)$ all consumers are indifferent and therefore there exists no equilibrium in which a mass at an atom are the only indifferent consumers.

We obtain the following Lemma, providing an alternative characterization for equilibria:

Lemma 3 (Equilibrium Characterization in Terms of Q_c^*). ***A.** The following are equivalent.*

***I.** There is a steady-state competitive equilibrium with $p_u^* \geq 0$, consistent externality impacts r_n^* and r_u^* giving rise to the cross-market effect $Q_c^* \in [0, (1 - f)]$, a cutoff type $k^* \in (\underline{k}, \bar{k})$ such that types $k < k^*$ buy new and types $k > k^*$ buy used, and consumption levels c_k for $k \in [\underline{k}, \bar{k}]$ given by Fact 1 with $p_u = p_u^*$.*

***II.** For (the same) p_u^* , Q_c^* , consumption levels c_k for $k \in [\underline{k}, \bar{k}]$ given by Fact 1 where types $k < k^*$ buy new and types $k > k^*$ buy used:*

$k^ \notin \mathcal{A}(G)$ and*

- 1. The cross-market effect Q_c^* satisfies Equation (5).*
- 2. The type k^* is indifferent: $f((1 - f) - Q_c^*)k^* = (p_u^* - (1 - f)P)(f + Q_c^*)$.*
- 3. The secondary market clears: $(1 - f) \int_{k \in [\underline{k}, k^*]} c_k dG(k) = f \int_{k \in [k^*, \bar{k}]} c_k dG(k)$.*

or $k^ \in \mathcal{A}(G)$ and*

¹⁹ A symmetrical argument holds when all indifferent consumers consume new goods.

1. The cross-market effect Q_c^* is $1 - f$.
2. The equilibrium price satisfies $p_u^* = (1 - f)P$ and all consumers are indifferent.
3. The secondary market clears for some $\alpha \in [0, 1]$ that determines the share of new products consumers with type $k = k^*$ buy:

$$(1 - f) \left(\int_{k \in [\underline{k}, k^*)} c_k dG(k) + \alpha c_{k^*} \Delta G(k^*) \right) = f \left(\int_{k \in (k^*, \bar{k}]} c_k dG(k) + (1 - \alpha) c_{k^*} \Delta G(k^*) \right)$$

B. Furthermore, any steady-state competitive equilibrium is payoff equivalent (for all consumer types as well as social welfare) to one in I.

Proof. **A.I** $\Rightarrow$ **II**. Take an equilibrium described in I. If $k^* \notin \mathcal{A}(G)$, then our above derivation establishes that Q_c^* satisfies Condition (5). Further, we have shown that either $Q_c^* = (1 - f)$ and $p_u^* = (1 - f)P$, or there is an indifferent consumer satisfies condition 2; but notice that for $Q_c^* = 1 - f$ and $p_u^* = (1 - f)P$, all types fulfill that condition, in which case k^* is determined by market clearing. Finally, since consumption levels are the same, market clearing is obvious.

If $k^* \in \mathcal{A}(G)$, then our above derivation establishes that the only equilibrium features $p_u^* = (1 - f)P$, $Q_c^* = (1 - f)$, and all consumers being indifferent. Finally, given that types $k < k^*$ buy new and types $k > k^*$ buy used, the mass at k^* has to split in a way that is consistent with market clearing which defines the α .

II $\Rightarrow$ **I**. First, consider a $k^* \notin \mathcal{A}(G)$. Define $r_n^* = \frac{f}{f + Q_c^*}$ and $r_u^* = Q_c^* r_n^*$. Since Q_c^* satisfies Condition (5), the impacts r_n^* and r_u^* generate a cross-market effect equal to Q_c^* and are consistent. Furthermore, fact 1 implies that if type k^* is indifferent, then types $k < k^*$ weakly prefer the new product and types $k > k^*$ weakly prefer the used one. Hence, consumer behavior is consistent with equilibrium. Market clearing holds by construction.

Consider now $k^* \in \mathcal{A}(G)$. Define $r_n^* = f$ and $r_u^* = (1 - f)f$. We have already showed that at $p_u^* = (1 - f)P$ these impacts are consistent. As all consumers are indifferent and consumption levels are given by fact 1, consumer behavior is optimal. Finally, market clearing holds by construction when consumers with k^* split their consumption according to α , and since they are indifferent this behavior is optimal.

B. Again from our derivation, any steady-state competitive equilibrium implies a $Q_c^* \in [0, 1 - f]$. Furthermore, if $Q_c^* < 1 - f$, then there is a unique type $k^* \notin \mathcal{A}(G)$ who is indifferent between new and used products, and types $k < k^*$ strictly prefer new, while types $k > k^*$ strictly prefer used. Hence, the equilibrium is described by I. If $Q_c^* = 1 - f$, then $p_u^* = (1 - f)P$, and all consumers

are indifferent between new and used products. In this case, each consumer's consumption level and utility are independent of which product she chooses, and so is production and social welfare. Hence, any such equilibrium is payoff-equivalent to one in which we choose k^* (potentially at an atom) such that types $k < k^*$ buy new and types $k > k^*$ buy used, and market clearing holds. $\square$

Proof of Proposition 3. I. Assume $Q_c^* = 1 - f$ and $p_u^* = (1 - f)P$. Now all consumers are indifferent, consumption levels are given by fact 1 and are independent of the type of good consumers choose. Denote the total consumption level by c_k . Consider the following auxiliary construction which tracks market clearing. We stretch the line $[\underline{k}, \bar{k}]$ by replacing any $k \in \mathcal{A}(G)$ with a unit interval, where the point on this interval tracks the fraction of indifferent consumers consumption that comes from new goods. Then consider the following function: at any point that corresponds to a $l \notin \mathcal{A}(G)$ it equals

$$(1 - f) \int_{k \in [\underline{k}, l]} c_k dG(k) - f \int_{k \in [l, \bar{k}]} c_k dG(k)$$

and at any point corresponding to an $l \in \mathcal{A}(G)$ and a point $\alpha \in [0, 1]$ on the associated unit interval it equals

$$(1 - f) \left(\int_{k \in [\underline{k}, k^*]} c_k dG(k) + \alpha c_{k^*} \Delta G(k^*) \right) - f \left(\int_{k \in (k^*, \bar{k}]} c_k dG(k) + (1 - \alpha) c_{k^*} \Delta G(k^*) \right).$$

This is a continuous strictly increasing function that starts below 0 and ends up above 0. Hence, there exists a unique point at which it equals 0. If this point corresponds to a $k \notin \mathcal{A}(G)$ then set $k^* = k$. Since for $k \notin \mathcal{A}(G)$ the first equilibrium condition is always fulfilled when $Q_c^* = (1 - f)$ and the second condition

$$f((1 - f) - Q_c^*)k^* = (p_u^* - (1 - f)P)(f + Q_c^*) \iff f0k^* = (p_u^* - (1 - f)P)1 \iff p_u^* = (1 - f)P$$

is independent of k^* , the conditions in part II of Lemma 3 are fulfilled and there exists an equilibrium as claimed. If the point corresponds to a $k \in \mathcal{A}(G)$, then $Q_c^* = 1 - f$ and $p_u^* = (1 - f)P$ by assumption and there exists a split of the mass (determined by the corresponding α) at k^* such that market clearing holds. Again, the conditions in part II of Lemma 3 are fulfilled and there exists an equilibrium as claimed.

Since all consumers are indifferent, their total consumption is as without a secondary market

by Lemma 1 and since all used goods are consumed with and without the secondary market, so is welfare.

II. In any other equilibrium $Q_c^* < (1 - f)$ so that $r_u^* < (1 - f)r_n^*$ and we are in the $k^* \notin \mathcal{A}(G)$ case. We also have that $(1 - f)P < p_u^* < P$. The upper bound holds, since otherwise demand of new goods would be infinite. The lower bound holds with weak inequality, since otherwise demand for new goods would be zero and with equality demand is positive only if $Q_c^* = (1 - f)$. There is also a unique indifferent consumer and all others strictly prefer one good. Therefore, the indifferent type must be $k^* \in (\underline{k}, \bar{k})$. Hence total consumption of almost all consumers is strictly higher than without a secondary market by Lemma 1. Hence, production is strictly higher by Lemma 2 and, since used goods are consumed either case, welfare is strictly lower than without a secondary market.

III. Let $\underline{g} = \inf\{g(x)|x \in \text{supp}(g)\}$.

In order to have an equilibrium with $Q_c^* < (1 - f)$, the following condition needs to hold:

$$Q_c^* = \frac{(1 - f)(f + Q_c^*)g(k^*)c_{k^*}}{(f + Q_c^*)g(k^*)c_{k^*} - f((1 - f) - Q_c^*) \int_{k^*}^{\bar{k}} (1/u''(c_k))dG(k)} = \frac{(1 - f)A}{A + f((1 - f) - Q_c^*)B}$$

Note that we moved the minus sign under the integral for convenience, so that both A and B are strictly positive. We will show that there is a $\underline{g}$ sufficiently large s.t. for every $Q_c^* < (1 - f)$, the RHS strictly exceeds the LHS. That is, we will show that $Q_c^* < \frac{(1-f)A}{A+f((1-f)-Q_c^*)B}$. Thus, assume that $Q_c^* < (1 - f)$, then we have:

$$\begin{aligned} Q_c^* < \frac{(1 - f)A}{A + f((1 - f) - Q_c^*)B} &\iff Q_c^*A + Q_c^*f((1 - f) - Q_c^*)B < (1 - f)A \\ &\iff Q_c^*f((1 - f) - Q_c^*)B < ((1 - f) - Q_c^*)A \iff Q_c^*fB < A \end{aligned}$$

where the last equivalence follows because we assume that $Q_c^* < (1 - f)$. Moreover, it is sufficient to show that $fB < A$. Spelling out A and B , we get that this is equivalent to

$$g(k^*) > \frac{f}{(f + Q_c^*)c_{k^*}} \int_{k^*}^{\bar{k}} (-1/u''(c_k))dG(k) \quad (6)$$

Note that, independent of $\underline{g}$, c_{k^*} is bounded below by $c_{\bar{k}}$ and $f + Q_c^*$ is bounded below by f , so the first LHS term is strictly bounded and we only need to bound the integral. Let $m = \max_{\underline{k}, \bar{k}} (-1/u''(c_k))$, which exists and is finite, since the function is continuous and c_k is bounded away from 0 for all k we consider, thus the term doesn't blow up. Hence the integral is strictly less than $\int_{\underline{k}}^{\bar{k}} m dG(k) = m$.

Thus the term on the right in equation (6) is bounded above, so if $\underline{g}$ exceeds this bound, we have no other equilibrium. This proves the first part of III.

For the second part, first assume that G has no atoms. Consider the following **claim**: For every $q \in [0, (1 - f)]$, there is a unique $k(q)$ and $p_u(q)$ such that the indifference condition for $k(q)$ and market clearing are satisfied.

Proof of claim: Let $p_u(q, k)$ be the unique solution to $f((1 - f) - q)k = (p_u(q, k) - (1 - f)P)(f + q)$. Then we have that $S(q, \underline{k}) = 0$ and $D(q, \underline{k}) = \frac{f}{1-f} \int_{\underline{k}}^{\bar{k}} c_k g(k) dk > 0$, as $c_k > 0$ for all k . Similarly, $S(q, \bar{k}) = \int_{\underline{k}}^{\bar{k}} c_k g(k) dk > 0 = D(q, \bar{k})$. Moreover, $S(q, k)$ is strictly increasing in k , while $D(q, k)$ is strictly decreasing in k : first, the range over which we integrate is increasing for S and decreasing for D . Further, $p_u(q, k)$ is strictly increasing in k . Hence for $k' < k$, we have that $c_{k'}$ is increasing in $p_u(q, k)$, hence the integral - i.e., S - is strictly increasing in k . When $k' > k$, then $c_{k'}$ is decreasing in $p_u(q, k)$, since consumers have to pay the price of the used good. Thus $D(q, k)$ is strictly decreasing in k . Hence there is a unique k s.t. $S(q, k) = D(q, k)$ holds for $p_u(q, k)$. This is $k(q)$, and $p_u(q) = p_u(q, k(q))$, proving the claim.

Now pick any $q \in (0, (1 - f))$. Then let $k_0 = k(0)$, and consider some $\varepsilon > 0$. We will construct $\hat{G}$ s.t. $\hat{k}(q) = k(q)$, $\hat{p}_u(q) = p_u(q)$, with $\hat{g}(k(q)) = \varepsilon$. To construct this $\hat{G}$, define

$$H_a(x, \varepsilon) = \begin{cases} 0 & \text{if } x \leq a - \varepsilon \\ 1 & \text{if } x \geq a + \varepsilon \\ \frac{1}{2\varepsilon}(x - a + \varepsilon) & \text{for } x \in (a - \varepsilon, a + \varepsilon) \end{cases}$$

Next we define $\hat{g}$ as follows, for some $\varepsilon_2 > 0$:

$$\hat{g}(x) = g(x) + (g(x) - \varepsilon) (H_{k(q)+\lambda\varepsilon}(x, \varepsilon_2) - H_{k(q)-\varepsilon}(x, \varepsilon_2))$$

We will deal later with the fact that $\hat{g}$ is not a probability density integrating to 1.

Except for buffer intervals of size ε_2 at the boundaries, this function equals $g(x)$ outside of $[k(q) - \varepsilon, k(q) + \lambda\varepsilon]$, and equals ε within. In what follows we denote with $R(\varepsilon_2)$ all terms coming from these buffers (this is a slight abuse of notation, since the terms vary). It is straightforward that as $\varepsilon_2 \rightarrow 0$ we have $R(\varepsilon_2) \rightarrow 0$. We pick λ s.t. market clearing holds under q , $k(q)$, and $p_u(q)$

with the new distribution. This is possible, since the new supply is

$$\hat{S}(q, k(q)) = \int_{\underline{k}}^{k(q)} c_k \hat{g}(k) dk = \left(\int_{\underline{k}}^{k(q)-\varepsilon} c_k g(k) dk + \int_{k(q)-\varepsilon}^{k(q)} \varepsilon c_k dk + R(\varepsilon_2) \right),$$

so that $S(q, k(q)) - \hat{S}(q, k(q)) = \int_{k(q)-\varepsilon}^{k(q)} c_k (g(k) - \varepsilon) dk + R(\varepsilon_2)$. As for the new demand, we have

$$\hat{D}(q, k(q)) = \frac{f}{1-f} \int_{k(q)}^{\bar{k}} c_k \hat{g}(k) dk = \frac{f}{1-f} \left(\int_{k(q)}^{k(q)+\lambda\varepsilon} c_k \varepsilon dk + \int_{k(q)+\lambda\varepsilon}^{\bar{k}} c_k g(k) dk + R(\varepsilon_2) \right)$$

so that $D(q, k(q)) - \hat{D}(q, k(q)) = \frac{f}{1-f} \int_{k(q)}^{k(q)+\lambda\varepsilon} c_k (g(k) - \varepsilon) dk + R(\varepsilon_2)$. Note that $D(q, k(q)) = S(q, k(q))$ by construction of $k(q)$ to satisfy market clearing, so

$$\hat{D}(q, k(q)) - \hat{S}(q, k(q)) = \int_{k(q)-\varepsilon}^{k(q)} c_k (g(k) - \varepsilon) dk - \frac{f}{1-f} \int_{k(q)}^{k(q)+\lambda\varepsilon} c_k (g(k) - \varepsilon) dk + R(\varepsilon_2).$$

For $\lambda = 0$, this difference is strictly positive for sufficiently small ε_2 (depending on ε). For sufficiently large λ , this difference is equal to $\int_{k(q)-\varepsilon}^{k(q)} c_k (g(k) - \varepsilon) dk - \frac{f}{1-f} \int_{k(q)}^{\bar{k}} c_k (g(k) - \varepsilon) dk + R(\varepsilon_2)$, with the terms depending on ε being strictly negative for sufficiently small ε .

Thus for all ε such that this holds, we can find ε_2 small enough and λ large enough so that the difference in demand and supply is given by the above, which is strictly negative. Hence for small enough ε_2 , the difference is strictly positive at $\lambda = 0$ and strictly negative at some large λ , and continuous in between, hence there is some $\lambda(\varepsilon)$ such that the difference is 0. Moreover, $\lambda(\varepsilon)\varepsilon \rightarrow 0$ as $\varepsilon \rightarrow 0$. If this was not the case, then $\hat{D}(q, k(q))$ would be strictly lower than $D(q, k(q))$ as ε and ε_2 converge to 0, yet $\hat{S}(q, k(q))$ converges to $S(q, k(q))$, so that $\lim \hat{D}(q, k(q)) - \hat{S}(q, k(q)) < 0$ as $\varepsilon \rightarrow 0$, even though $\hat{D}(q, k(q)) = \hat{S}(q, k(q))$ for all $\varepsilon > 0$. This is a contradiction.

The indifference condition for $k(q)$ is not affected by changes in the distribution and since $\lambda(\varepsilon)$ is s.t. markets clear. Thus we have a new density function $\hat{g}$ s.t. the indifference condition for $k(q)$ and market clearing hold for the old values of $k(q)$ and $p_u(q)$. Moreover $\hat{g}(k(q)) = \varepsilon$.

As mentioned above, this $\hat{g}$ does not integrate to 1, but we can scale $\hat{g}$ by the same factor everywhere, to obtain $a^{-1} \cdot \hat{g}(k)$, where $a = \int_{\underline{k}}^{\bar{k}} \hat{g}(k) dk$, with $a \rightarrow 1$ as $\varepsilon \rightarrow 0$. Let $\hat{h}(k) = a^{-1} \hat{g}(k)$. Then the indifference condition and market clearing hold under $\hat{h}$ with the old values of $k(q)$ and $p_u(q)$; the first since it doesn't depend on the distribution, the second since market clearing isn't affected by a multiplicative rescaling of the distribution.

Now we are ready to consider the final equilibrium condition which determines Q_c^* , but to do

so for $\hat{h}$ instead of g :

$$Q_c^*(q) \equiv (1-f) \left[\frac{(f+q)\hat{h}(k(q))c_{k(q)}}{(f+q)\hat{h}(k(q))c_{k(q)} - f((1-f)-q) \int_{k^*}^{\bar{k}} (1/u''(c_k))\hat{h}(k)dk} \right]$$

Note that if we find a q s.t. $Q_c^*(q) = q$, then for this q we have that all the equilibrium conditions hold, and hence we have identified an equilibrium.

It is clear that $Q_c^*(0) > 0$ by plugging in $q = 0$. Moreover, plugging in the value of $q \in (0, (1-f))$ for which we constructed $\hat{h}$, we know that $\hat{g}(k(q)) = \varepsilon$, hence we have

$$Q_c^*(q) \equiv (1-f) \left[\frac{(f+q)\varepsilon c_{k(q)}}{(f+q)\varepsilon c_{k(q)} - f((1-f)-q) \int_{k^*}^{\bar{k}} (1/u''(c_k))\hat{h}(k)dk} \right]$$

The second term in the denominator, $-f((1-f)-q) \int_{k^*}^{\bar{k}} (1/u''(c_k))\hat{h}(k)dk$ is strictly positive and strictly bounded away from 0, since market clearing requires that $k(q)$ cannot equal $\bar{k}$. Hence $Q_c^*(q) \rightarrow 0$ as $\varepsilon \rightarrow 0$, hence there is $\varepsilon > 0$ s.t. $Q_c^*(q) < q$.

But then, since $Q_c^*(q)$ is continuous in q , there must be some $q' \in (0, q)$ s.t. $Q_c^*(q') = q'$, and thus we have an equilibrium with $Q_c^* < (1-f)$. Finally note that $\hat{h}$ and g only differ by an arbitrary small amount except on an arbitrarily small range (where they differ a discrete amount). Hence, a suitable $\hat{H}$ can be chosen arbitrarily close to G in the L_1 -norm. Note that since q was arbitrary, we can pick it arbitrarily small, so that we can ensure that there is such an equilibrium with q' , and hence Q_c^* , arbitrarily small.

Now for any G with atoms we can first replace it with an, in the L_1 -norm, arbitrarily close distribution without atoms and then do the above construction. $\square$

Proof of Proposition 4. We first prove that there exist distributions where consumption in the best equilibrium differs by at most ϵ and in the worst equilibrium by at least W , where ϵ is arbitrarily small and W arbitrary large. Moreover, instead of working with the utility function $u(\cdot)$ we will construct the policy function $C(\cdot)$. On any set bounded away from zero and infinity, we can freely choose any function that is strictly decreasing, continuously differentiable, and it will give rise to a utility function satisfying our assumptions.

Now, pick any $\bar{k}$ and $\underline{k}$ with $K > \bar{k} > \underline{k} > 0$ and let δ be (arbitrarily) close to zero. Consider the following $C(\cdot)$: on $[f(P + \underline{k} - \delta), f(P + \bar{k})]$ it is linearly decreasing with $C(f(P + \underline{k} - \delta)) = c$ and $C(f(P + \bar{k})) = c - \varepsilon$; $C(f(P + \underline{k} - 2\delta)) = c^+ > c$; and this is extended smoothly and strictly

decreasing on $[f(P + \underline{k} - 2\delta), f(P + \underline{k} - \delta)]$.

Let $\hat{G}_h$ be uniformly distributed on $[\underline{k}, \bar{k}]$. Then, let G_h be the distribution (as constructed in the proof of Part III of Proposition 3) that is arbitrarily near to $\hat{G}_h$ in the sup norm, has the same support as $\hat{G}_h$, and for which an equilibrium with Q_c^* arbitrarily close to 0 exists. Let G_l be the uniform distribution on $[\underline{k} - \hat{\delta}, \underline{k}]$ where $\hat{\delta} < \delta$ and, as is possible by Part III of Proposition 3, sufficiently small that only the equilibrium with $Q_c^* = (1 - f)$ exists. It is clear that G_h first-order stochastically dominates G_l and that (i) holds. In the unique equilibrium under G_l consumers face an effective price of $f(P + k)$, where $k \in [\underline{k} - \hat{\delta}, \underline{k}]$. By construction of $C(\cdot)$, overall consumption is at most c .

In the best equilibrium under G_h everyone faces an effective price of $f(P + k)$, where $k \in [\underline{k}, \bar{k}]$. By construction, overall consumption is at least $c - \varepsilon$ and therefore within ε of the equilibrium consumption under G_l .

Consider the equilibrium under G_h with Q_c^* arbitrarily close to 0. In this equilibrium there is an indifferent type $k^* \in (\underline{k}, \bar{k})$, more selfish types prefer strictly new goods and consume $C(P + k - p_u^* - \varepsilon_1 k)$, and more responsible types prefer strictly used and consume $C(\varepsilon_2 k + p_u^* f / (1 - f))$. Notice that the social coefficient enters by at most ε_1 and ε_2 , which can be arbitrarily small, by picking Q_c^* arbitrarily close to 0.

Let $y = (1 - f/2)\underline{k} + f/2\bar{k}$ — i.e., a fraction $1 - f/2$ of types are above y and a fraction $f/2$ of types are below y (by near-uniform distribution of G_h). Then if $k^* \leq y$, a fraction $1 - f/2$ of types buy only used. Note that since the indifferent type consumes as much as without a secondary market (by Lemma 1), they face an effective price that is as if without secondary markets, i.e., $f(P + k^*)$. Since consumers with $k > k^*$ buy only used, they face the same effective price plus $\varepsilon_2(k - k^*)$. Hence, they all consume more than $c - \varepsilon'_2$, where ε'_2 goes to zero as $\varepsilon_2 \rightarrow 0$ and $\varepsilon \rightarrow 0$. Aggregate used consumption is therefore at least $(1 - f/2)(c - \varepsilon')$. Total consumption equals $1/(1 - f)$ of the aggregate used consumption, thus is at least $\frac{(1-f/2)}{1-f}c - \varepsilon''_2$ (where $\varepsilon''_2 \rightarrow 0$ as $\varepsilon_2 \rightarrow 0$ and $\varepsilon \rightarrow 0$). The difference between it and c is at least $\frac{f}{2(1-f)}c - \varepsilon''_2$. Pick $\Delta = f/4(1 - f)$ so that for any ε_2 and ε small enough (as well as G_h near enough to $\hat{G}_h$), this difference is at least Δc .

If instead $k^* > y$, then all types with $k < k^*$ consume new, at an amount $C(P + k - p_u^* - \varepsilon_1 k)$. We know that $p_u^* \approx (1 - f)(P + k^*) > (1 - f)(P + y)$. The effective price of these consumers is $P + k - p_u^* - \varepsilon_1 k < P + k - p_u^*$, and $P + k - p_u^* < f(P + \underline{k}) \iff P + k - (1 - f)(P + k^*) < f(P + \underline{k}) \iff k < f\underline{k} + (1 - f)k^*$. Hence, if k is within at most $1 - f$ of the way between $\underline{k}$

and k^* , the effective price is lower than $f(P + \underline{k})$. Notice that the effective price is linear in k with coefficient close to 1. Hence, it is lower than $f(P + \underline{k} - 2\delta)$ if k is within at most $1 - f - \gamma$ of the way between $\underline{k}$ and k^* for a small γ ($\gamma \rightarrow 0$ as $\delta \rightarrow 0$). Since k is nearly uniformly distributed on this range, this corresponds to almost a fraction $1 - f$ of types below k^* , which is at least a fraction of $1 - f$ of types below y , i.e., at least $(1 - f)f/2 - \gamma'$ of total types. For sufficiently small δ this is greater than $(1 - f)f/4$. All of these types consume at least c^+ and thus the amount of new consumption is at least $(1 - f) \cdot f/4 \cdot c^+$. Overall consumption is $1/f$ times new consumption and, therefore, overall consumption is at least $(1 - f)/4 \cdot c^+$. Now pick c^+ such that this amount is greater than $(\Delta + 1)c$. Again, overall consumption is at least Δc greater than c .

Since c was arbitrary and Δ independent of c , we can choose it such that $\Delta c > W$. Then, the above shows that in the bad equilibrium consumption is at least W greater than c , which is greater than consumption under the equilibrium in G_l .

Now consider welfare. Notice that in all cases the gain of a unit of additional (compared to a situation without a secondary market) production is at most the additional consumption utility to the type consuming the least. Their benefit is at most the marginal utility at the initial point, which equals their effective price $f(P + \bar{k})$ (independent of all constructions after fixing $\bar{k}$). The social cost of producing equals $(P + K) > f(P + \bar{k})$. Therefore, any extra production lowers welfare by at least that amount, which is strictly positive. Similarly, the maximum welfare loss can be bounded in terms of production. Finally, Lemma 2 shows that production is proportional to consumption and hence the above statement about the comparison of consumption translates into a comparison in welfare. $\square$

Proof of Proposition 5. We prove that $Q_c^* < 1 - f$ requires $f \geq G(0)$. This implies that for $f < G(0)$, we have $Q_c^* = 1 - f$ and therefore $r_n^* = f$ and $r_u^* = (1 - f)f$, i.e., the type-I equilibrium obtains.

First, by Fact 1 the total consumption levels c_k in any equilibrium are weakly decreasing in k . Since in any equilibrium with $Q_c^* < (1 - f)$ the indifferent consumer cannot be at an atom, consumers at $k = 0$ have to strictly prefer new. Supply for used goods is $\int_{k \in [\underline{k}, k^*]} c_k dG(k) \geq c_0 G(0)$ and demand is $f/(1 - f) \int_{k \in [k^*, \bar{k}]} c_k dG(k) \leq f/(1 - f)(1 - G(k^*))c_{k^*} \leq f/(1 - f)(1 - G(0))c_0$. Market clearing thus requires that $c_0 G(0) \leq f/(1 - f)(1 - G(0))c_0 \implies \frac{G(0)}{1 - G(0)} \leq \frac{f}{1 - f}$. $\square$

Proof of Proposition 6. Consider a candidate equilibrium with $p_u^* = 0$, $r_u^* = 0$, and $r_n^* = 1$. If demand of used equals supply, then a change in used demand by Δ requires that some consumers

switch their demand from used to new given the fixed supply of used, hence $r_u^* > 0$. Thus this can be an equilibrium only if there is a strict excess supply of used goods. In this case, an increase in used demand can be absorbed without a price shift, so that it doesn't affect the consumption level of any other consumer. Thus production is only affected if there is a change in the source of indifferent consumers consumption (new vs. used). This change can only occur if a mass of consumers are indifferent and in this case, as before, no equilibrium exists. If there is no mass at the indifferent type, $r_u^* = 0$ and $r_n^* = 1$ are consistent. This shows that we have an equilibrium if and only if we have oversupply of used goods and no atom at the indifferent k^* . A consumer chooses new goods if $P + k < \frac{l}{(1-f)} \iff k < \frac{l}{1-f} - P$ and used if the inequality is strict in the other direction. Consumption levels are given by Fact 1: $u'^{-1}(P+k)$ for new and $(1-f)^{-1}u'^{-1}(l/(1-f))$ for used consumption. Since the second group can cover a share $1-f$ of their demand for used via their own consumption in the past period, we require that the consumption of new goods strictly exceeds f times the total amount of used goods demanded; i.e., there is an oversupply of used goods if and only if $\int_k^{l/(1-f)-P} u'^{-1}(P+k)dG(k) > \frac{f}{1-f} \cdot \left(1 - G(l/(1-f) - P)\right) u'^{-1}(l/(1-f))$. Therefore, this condition and no atom at $l/(1-f) - P$ is necessary and sufficient for the existence of the candidate equilibrium. Notice that all consumers that buy new buy the same amount of new goods as without a secondary market. Consumers buying used buy 0 new goods and hence strictly less new goods than without a secondary market. Therefore, total consumption of new goods and hence production is strictly lower, if there is a mass of consumers that buy used. Finally, by Lemma 1 all consumers consumption is weakly higher and hence welfare is strictly higher. $\square$

Proof of Proposition 7. Proposition 2 part III. shows that there are no strict equilibria. Consider now equilibria where at least one consumer type is indifferent. Since $p_u^* \geq 0$, selfish consumers always strictly prefer new goods, so that socially responsible consumers must be indifferent for a possible non-strict equilibrium. Then following a shift in demand, the new market-balancing price p_u required for consistency (page 16) must equal p_u^* : if it would change, indifference would be broken, which would change demand non-continuously. Since we require that the impacts arise as a differential, this would not be consistent. Therefore $p_u = p_u^*$ following the shift and total consumption by all consumers cannot change. Hence an increase of one unit $(1-f)$ units rescaled in used consumption would cause an increase of new consumption, and thus production, by $(1-f)$ units. The only potentially consistent impacts r_n^* and r_u^* are such that $r_u^* = (1-f)r_n^*$, so that the per-consumption-unit impact would be the same for used and new products. But then, because

$l > (1-f)P$, socially responsible consumers would strictly prefer new goods, a contradiction. Thus, there can't be any equilibrium with indifferent consumers and hence no equilibrium at all. $\square$

Proof of Proposition 8. Similarly to Fact 1 we can determine optimal consumer behavior by considering the effective price of a good and the policy function $C(\cdot)$. Since naive consumers only consider the direct effect, their effective prices are $P - p_u^* + k$ and $(1-f)^{-1}p_u^* - p_u^*$ for new and used goods, respectively. If $k < (1-f)^{-1}p_u^* - P$, a consumer prefers new products and consumes $C(P - p_u^* + k)$ units. If $k > (1-f)^{-1}p_u^* - P$, then she prefers used and consumes $(1-f)^{-1}C(\frac{f}{(1-f)}p_u^*)$ units. If $k = (1-f)^{-1}p_u^* - P$, the consumer is indifferent and her total consumption is $C(f(P+k))$. Notice that the only variable influencing consumers optimal consumption level is p_u^* . When the indifferent type is not at an atom then also overall supply and demand of used goods only depend on p_u^* . When there is an atom at $k = (1-f)^{-1}p_u^* - P$, demand and supply of used goods also depend on how indifferent consumers split their consumption between used and new goods. Consider now the demand and supply for used goods at a price p_u^* . To analyze market clearing we want to consider demand and supply as functions of p_u^* . As they are not identified when there is a mass of indifferent types, we again consider the auxiliary construction where replace any price that corresponds to an indifferent type at an atom by a unit interval that tracks the fraction of new goods the indifferent consumers consume. Now increasing p_u^* , and where required the fraction of new consumption by the indifferent mass, continuously changes demand and supply. Moreover, demand is clearly strictly decreasing and supply is clearly strictly increasing in price and the fraction purchasing new. At $p_u^* = (1-f)P$ (and if there is an atom, all purchasing used goods), supply is zero while demand is positive. At $p_u^* = (1-f)(P + \bar{k})$ (and if there is an atom, all purchasing new goods), supply is positive and demand is zero. By continuity and monotonicity, there therefore exists a unique price p_u^* (and potentially a fraction that determines how many consumers at an atom buy new) such that the used-goods market clears and consumers behave optimally.

Without a secondary market the consumption of type k is determined by the effective price $f(P+k)$. Assuming $k < (1-f)^{-1}p_u^* - P$ the consumer buys new and her effective price is $P - p_u^* + k < P - (1-f)(k+P) + k = f(P+k)$. If $k > (1-f)^{-1}p_u^* - P$, she buys used and faces the effective price $\frac{f}{(1-f)}p_u^* < f(P+k)$. In both cases the consumers consumes strictly more. If $k = (1-f)^{-1}p_u^* - P$, then she faces an effective price of $f p_u^* = f(P+k)$ and consumes exactly as much as without a secondary market. Thus, all consumer consume weakly more and almost all types consume strictly more. As in Lemma 2 the production is thus strictly higher which implies

that welfare is strictly lower than without a secondary market. $\square$

Proof of Observation 2. Suppose there exists a strict equilibrium. High-income, socially responsible consumers strictly prefer used goods iff $(1 - f)(P + \kappa) - l > p_u^*$. Low-income consumers strictly prefer used goods iff $(1 - f)P > p_u^*$. Recall that we assumed $l - (1 - f)\kappa < 0$ and hence, if $p_u^* > (1 - f)(P + \kappa) - l$, no consumer buys used goods which contradicts $p_u^* > 0$; if $p_u^* < (1 - f)P$, both strictly prefer used, in which case there is no supply of used goods and thus no equilibrium. Therefore, $(1 - f)P < p_u^* < (1 - f)P + \kappa(1 - f) - l$, and low-income, selfish consumers strictly prefer new and high-income, socially responsible consumers strictly prefer used goods. Since all consumers strictly prefer one type of good, consumption is strictly higher than without a secondary market by an identical argument as in Lemma 1²⁰. Since additionally $l < (1 - f)P$, production is strictly higher by an identical argument as in Lemma 2²¹. Moreover, the allocation efficiency is lower because used goods are only consumed by high-income consumers who receive additional disutility l while without a secondary market all consumers consume used goods. Since production is higher while allocation efficiency is lower, welfare is strictly lower than without a secondary market. $\square$

D A Small Consumer’s Impact on Production

In this section, we restate the main insight in Section II.A of Kaufmann et al. (2024) in a different way. This motivates our definition of a consumer’s impact in Section 2.

For simplicity, we use Kaufmann et al.’s static framework; adapting the analysis to our dynamic setting is notationally heavy, but does not change the logic. To understand a small consumer’s impact, we use a “replicator economy:” we introduce identical copies of the other participants, and let the number of copies approach infinity. Suppose, then, that our single individual enters a market with I other consumers and I suppliers. The other consumers all have the same demand curve $D(p)$, and the suppliers all have the same supply curve $S(p)$. Both curves are continuously differentiable, with $D'(p) < 0$ and $S'(p) > 0$ everywhere. There is a price $p^* > 0$ for which $S(p^*) = D(p^*)$.

The market mechanism is the following. First, the individual submits her demand $c \in \mathbb{R}$.

²⁰While we have formulated Lemma 1 for an equilibrium with homogeneous l , the argument is based on individual consumers effective prices. Here, the heterogeneous l affects the effective prices but for each consumer type the same argument holds.

²¹Again, we formulated it for an equilibrium with homogeneous l , but individual consumption aggregates the same in this setting and is related to overall production in the same way.

Then, the price $p_I(c) > 0$ is chosen to clear the market, i.e., to satisfy $c + ID(p_I(c)) = IS(p_I(c))$; suppose that $p_I(c)$ exists and is unique for all I and c of interest. Finally, the equilibrium quantity $q(c) = IS(p_I(c))$ is produced and consumed.

A vanishingly small consumer's impact on the price is then obviously zero: for any c , $\lim_{I \rightarrow \infty} p(c) = p^*$. For her impact on production, note that $c + ID(p_I(c)) = IS(p_I(c))$ is equivalent to $c/I + D(p_I(c)) = S(p_I(c))$, so that $p_I(c) = p_1(c/I)$. Hence, the individual's impact is

$$IS(p_I(c)) - IS(p_I(0)) = \frac{S(p_1(c/I)) - S(p_1(0))}{1/I} = c \cdot \frac{S(p_1(c/I)) - S(p_1(0))}{c/I}$$

Taking the limit yields that a vanishingly small consumer's impact is

$$c \cdot \lim_{I \rightarrow \infty} \frac{S(p_1(c/I)) - S(p_1(0))}{c/I} = c \cdot r^*,$$

where r^* is the marginal effect of shifting demand in an economy with representative demand curve $D(\cdot)$ and supply curve $S(\cdot)$ (i.e., the derivative of production with respect to Δ at $\Delta = 0$ when the demand curve is $\Delta + D(p)$ and the supply curve is $S(p)$). Our definition of r_n^* and r_u^* adapts this observation to our more complicated setting.